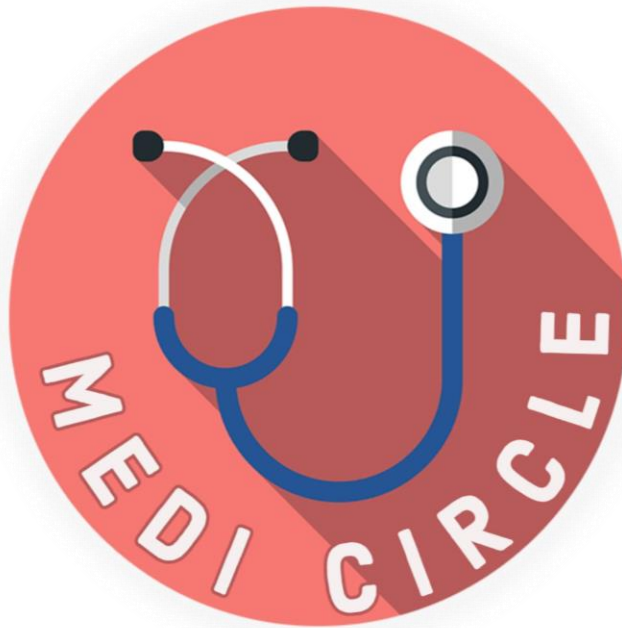


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**FIRST YEAR SIA Q BANKS, EXAM ORIENTED PDFS,  
TEXT BOOKS**

**<https://t.me/medicirclebooks1>**

KALOJI NARAYANA RAO UNIVERSITY OF HEALTH SCIENCES  
WARANGAL, TELANGANA STATE – 506 002

MBBS FIRST YEAR EXAMINATIONS: NOVEMBER, 2023

**BIOCHEMISTRY (New Regulation)**  
**PAPER – I**

Time: 3 Hours

Max Marks: 100

Note: Answer all questions

Draw diagrams whenever necessary

**KNRUHS -2022**

Use HB pencil /any dark

Colour pencil

**Multiple Choice Questions:**

**10 X 1 = 10**

1. Anomeric carbon atom of fructose is in
  - a) 1<sup>st</sup> carbon atom
  - b) 2<sup>nd</sup> carbon atom
  - c) Penultimate carbon atom
  - d) None of the above
2. All the following are unsaturated fatty acids except
  - a) Propionic acid
  - b) Linoleic acid
  - c) Linolenic acid
  - d) Arachidonic acid
3. During glycogen breakdown, the action of glycogen phosphorylase produces
  - a) Glucose -6-phosphate
  - b) Glucose-1-phosphate
  - c) Fructose- 6-phosphate
  - d) None of the above
4. The key enzyme in the pathway of cholesterol biosynthesis is
  - a) HMG CoA synthase
  - b) HMG CoA lyase
  - c) HMG CoA reductase
  - d) Mevalonate kinase
5. Components of the electron transport chain are arranged in order of
  - a) Decreasing redox potential
  - b) Increasing redox potential
  - c) Positive redox potential to negative redox potential
  - d) None of the above
6. Enzymes act by reducing the
  - a) Activation energy
  - b) Heat energy
  - c) Binding energy
  - d) All of the above
7. Increased levels of lactate dehydrogenase (LDH) can be observed in
  - a) Myocardial infarction
  - b) Haemolytic anaemia
  - c) Leukaemia's
  - d) All of the above

@medi\_circle

8. The fluid mosaic model of membrane includes
- |                                                                         |                                                                                         |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| a) Cholesterol molecules in the membrane increase stability of membrane | b) Membrane proteins are fixed and cannot move                                          |
| c) Membrane contains bimolecular leaflet of lipids                      | d) Permeability of drugs across cell membrane is dependent on their solubility in water |
9. Mobility of particles during electrophoresis is based on all the following factors, except
- |                               |                                   |
|-------------------------------|-----------------------------------|
| a) Current voltage and ampere | b) Quality of supporting media    |
| c) Ph of buffer medium        | d) Solubility of particle protein |
10. The power house of the cell is
- |                 |                  |
|-----------------|------------------|
| a) Nucleus      | b) Cell membrane |
| c) Mitochondria | d) Lysosomes     |

### Essey/Long Answer Questions:

2 X 15 = 30

11. What are Lipoproteins. Classify Lipoproteins. Explain the transport of TAG (Tri Acyl glycerides) in the body. Discuss the clinical significance of LDL and HDL. Add a note on dyslipidaemias. (2+3+5+2+3)
12. What is Glycogen. How is Glycogen broken down in the body. Explain the hormonal regulation of the pathway. Write about glycogen storage disorders. (1+4+4+6)

### Short Answer Questions:

**@medi\_circle**

7 X 6 = 42

13. What are the different types of enzyme inhibition. Explain about competitive inhibition with two examples of its therapeutic significance.
14. Write about inhibitors and uncouplers of electron transport chain.
15. Justify Physician role in Health Care System.
16. Enzyme profile in liver diseases, and write about their clinical significance.
17. Describe the sources, daily requirement and functions of Vitamin C.
18. Describe the regulation of calcium homeostasis.
19. Define BMR, what is its normal value, and what are all the factors affecting BMR.

### Very Short Answer Questions:

6 X 3 = 18

20. Write about classification of lipids with suitable examples.
21. Principle and applications of Enzyme Linked Immunosorbent Assay (ELISA).
22. Antioxidant Vitamins.
23. Describe about transport systems with examples.
24. Essential Fatty acids.
25. Compare and Contrast Kwashiorkor and Marasmus.

**KALOJI NARAYANA RAO UNIVERSITY OF HEALTH SCIENCES**

**WARANGAL, TELANGANA STATE – 506 002**

**MBBS FIRST YEAR EXAMINATIONS: NOVEMBER, 2023**

**BIOCHEMISTRY (New Regulation)**

**PAPER – II**

**Time: 3 Hours**

**Max Marks: 100**

**Note:** Answer all questions

Draw diagrams whenever necessary with Black Ball point pen /HB pencil /any dark

Colour pencil

**Multiple Choice Questions:**

**10 X 1 = 10**

1. Peptide bond is
  - a) Single bond in character
  - b) Double bond in character ✓
  - c) Partial double bond in character
  - d) Triple bond in character
2. During urea cycle, the two nitrogen atoms are derived from
  - a) Ammonia and Arginine
  - b) Ammonia and Aspartic acid ✓
  - c) Both from Ammonia
  - d) Ammonia and Ornithine
3. The sources of oxalic acid in urine are
  - a) Ornithine and citrulline
  - b) Oxalosuccinate and formic acid
  - c) Oxaloacetate and aspartic acid
  - d) Ascorbic acid and Glycine ✓
4. Elevated homocysteine levels are associated with a risk of
  - a) Bronchiectasis
  - b) Nephrotic syndrome
  - c) Coronary artery disease
  - d) Diabetes mellitus
5. Regarding DNA structure which is not correct.
  - a) Two strands are covalently bonded
  - b) Two strands are antiparallel
  - c) Strands have polarity ✓
  - d) Strands are complementary to each other
6. Which of the following conditions is not associated with hyperuricemia.
  - a) Multiple myeloma
  - b) Xanthinuria ✓
  - c) Psoriasis ✓
  - d) Polycythemia vera
7. Sickle cell anaemia occurs due to
  - a) Silent mutation
  - b) Mis-sense mutation
  - c) Nonsense mutation
  - d) Frameshift mutation ✓
8. Which of the following renal functional tests are used to measure the Glomerular Filtration Rate (GFR)
  - a) Clearance tests ✓
  - b) Urine concentration tests
  - c) Urine dilution tests
  - d) Urine acidification tests
9. Which is an anticancer drug.
  - a) Zidovudine
  - b) 6-mercaptopurine
  - c) Allopurinol
  - d) Acyclovir

10. Okazaki fragments occur on the
- |                              |                   |
|------------------------------|-------------------|
| a) Leading strand            | b) Lagging strand |
| c) Double helical DNA strand | d) All the above  |

**Essay/ Long Answer Questions:**

**2 X 15 = 30**

11. A 15-year-old boy complained of swelling and pain in the distal phalangeal joints. Blood investigations showed the following results. On diagnosing the pathology, the physician decided to treat patient with allopurinol.
- What is your probable diagnosis in the above patient.
  - Write the synthesis of Uric Acid.
  - What is the normal uric acid level.  $< 3 \text{ mg/dl}$
  - What are other blood investigations would you suggest.
  - Comment on the serum uric acid level and explain the cause of pain in joints.
  - Write the ways for lowering serum uric acid level.
  - What is the biochemical explanation for the treatment given in the above patient. (1+3+2+3+3+3)
12. Describe the steps of catabolism of Phenylalanine and Tyrosine. Indicate the inborn errors of metabolism. What are the Biomedically important compounds derived from tyrosine. (8+5+2)

**Short Answer Questions:**

**Medi Circle**

**7 X 6 = 42**

13. Name the enzymes measured to assess liver functions. Discuss the clinical significance of Alanine transaminase and Alkaline phosphatase in assessing liver function.
14. Discuss the Watson and Crick model of DNA. Add a note on different forms of DNA.
15. Write on the classification of Acid-base disorders in the body and also mention the compensatory mechanisms for each type.
16. Justify the statement "Physician is a lifelong learner".
17. Describe the formation of bilirubin. Name the different types of jaundice.
18. What are tumour markers. Discuss the clinical utility of any three tumour markers.
19. Write about structure, classification and functions of immunoglobulins.

**Very Short Answer Questions:**

**6 X 3 = 18**

20. Orotic aciduria.
21. List the thyroid function tests. Mention the laboratory findings in a case of hypothyroidism.
22. What is Polymerase Chain Reaction (PCR). Mention two clinical applications of PCR technique.
23. Importance of cytochrome P450 in detoxification.
24. What is foetal haemoglobin. How does its structure differ from adult haemoglobin and explain how it is useful to the foetus.
25. ACTH.

# BIOCHEMISTRY (New Regulation)

## PAPER - I

Time: 3 Hours

Note: Answer all questions

Draw diagrams whenever necessary with Black Ball point pen /HB pencil /any dark

Colour pencil

Max Marks: 100

### Multiple Choice Questions:

10 X 1 = 10

- Volume expander in treatment of hypovolemia
  - Dextrins
  - ~~b) Dextran~~
  - Heparin
  - Agarose
- Pyruvate is converted to acetyl CoA by
  - ~~a) Pyruvate dehydrogenase~~
  - Pyruvate Carboxylase
  - Pyruvate kinase
  - Lactate dehydrogenase
- The presence of the following fatty acid in diet prevents heart attacks
  - Eicosa trienoic acid
  - Eicosa tetraenoic acid
  - ~~c) Eicosa pentaenoic acid~~
  - None of the above
- Prostaglandins are stored
  - In adipose tissue
  - As plasma lipoproteins
  - ~~a) As membrane phospholipids~~
  - As granules in mast cells
- All the following enzyme complexes of respiratory chain act as proton pumps except
  - Complex I
  - ~~b) Complex II~~
  - Complex III
  - Complex IV
- The enzyme involved in the transfer of electrons are known as
  - ~~a) Oxidoreductases~~
  - Transferases
  - Lyases
  - All of the above
- Symport system operates in the transport of
  - Bicarbonate
  - Chloride
  - Calcium
  - ~~d) Glucose~~
- The following diseases are peroxisomal biogenesis disorders except
  - Adrenoleukodystrophy
  - Zellweger syndrome
  - Primary Hyperoxaluria
  - Inclusion cell disease
- The separation of molecules based on their size, shape and molecular weight is done by
  - Ion-exchange Chromatography
  - Gel filtration chromatography
  - ~~a) Adsorption chromatography~~
  - Affinity chromatography

10. All the following coenzymes are derivatives of B-complex vitamins, except

a) S-adenosyl methionine

b) NAD<sup>+</sup>

c) Pyridoxal phosphate

d) Coenzyme A

#### Essay/ Long Answer Questions:

2 X 15 = 30

11. How the Fatty acids are oxidized in the body. Explain  $\beta$ -oxidation, give its energetics and explain the regulation of  $\beta$ -oxidation in the body.

(2+7+3+3)

12. An 8-year-old child was brought to the hospital with c/o slow growth and pain in bones. On examination, he was anaemic had frontal bossing, bowing of legs and swelling of costochondral junction. Laboratory results were serum calcium – 8.2mg/dl, serum ALP: 720 U/L, serum phosphorous – 2.8mg/dl.

a) What is your diagnosis.

b) What are the biochemical effects of vitamin D.

c) Causes for vitamin D deficiency.

d) Sources and RDA vitamin D.

e) What are the clinical features of vitamin D deficiency.

#### Short Answer Questions:

7 X 6 = 42

13. Describe the Hexose Mono Phosphate shunt path way and add a note on its significance.

14. What is Substrate Level Phosphorylation and explain with suitable examples.

15. Clinically important enzymes and their significance.

16. Mucopolysaccharides.

17. Write about Anti-oxidants and their role in health.

18. How does the Physician improves the Health Care System.

19. Absorption of iron and its regulation.

#### Very Short Answer Questions:

6 X 3 = 18

20. Dietary fiber.

21. What is Michaelis Menten ( $K_m$ ) value. Explain the concept by using suitable example.

22. Glycosuria and Renal Glycosuria.

(1+2)

23. Structure and functions of Lipoprotein.

24. Rothera's test.

25. Mention the clinical features of the following diseased conditions;

a. Keshan disease

b. Wilson's disease

c. Hemosiderosis.

**BIOCHEMISTRY (New Regulation)**  
**PAPER – II**

Time: 3 Hours

Max Marks: 100

Note: Answer all questions

Draw diagrams whenever necessary with Black Ball point pen /HB pencil /any dark  
Colour pencil

**Multiple Choice Questions:**

10 X 1 = 10

1. A Zwitter ion is
  - a) Positive
  - b) Negative
  - c) No net charge
  - d) None of the above
2. The major donor of carbon atom to the one carbon pool is
  - a) Serine
  - b) Tyrosine
  - c) Threonine
  - d) Proline
3. Which enzyme protects DNA from ageing.
  - a) DNA polymerase
  - b) Topoisomerase
  - c) Deoxyribonuclease
  - d) Telomerase
4. Which amino acid is required for both purine and pyrimidine synthesis.
  - a) Glycine
  - b) Aspartate
  - c) Alanine
  - d) Glutamine
5. Which one of the following is a termination codon in mitochondrial translation
  - a) UAA
  - b) UAG
  - c) UGA
  - d) AGA
6. Tumour marker for prostate cancer are
  - a) Alkaline phosphatase
  - b) Prostate specific antigen
  - c) Bence-Jones proteins
  - d) Calcitonin
7. Which of the following diseases are not attributed to free radical damage.
  - a) Rheumatoid arthritis
  - b) Chronic glomerulonephritis
  - c) Hypothyroidism
  - d) Cataract
8. The predominant cation in extracellular fluid (ECF) is
  - a) Na
  - b) K
  - c) Ca
  - d) Mg

9. At the pH of blood 7.4, the ratio between the carbonic acid and bicarbonate fractions is

- a) 1 : 10
- b) 1 : 20
- c) 1 : 30
- d) 1 : 40

10. Monoclonal antibodies are useful for

- a) Early detection of pregnancy
- b) Treatment of cancer
- c) Treatment of autoimmune diseases
- d) All the above

### Essay/ Long Answer Questions:

2 X 15 = 30

11. Why is ammonia toxic to the body. Describe the metabolic reactions involved in the detoxification of ammonia. Add a note on the defects in the metabolic cycle associated with detoxification of ammonia.

(4+7+4)

12. What is the normal blood  $P^H$ . Explain the role of plasma buffers and renal mechanism in the maintenance of acid base balance of the body.

(1+7+7)

### Short Answer Questions:

7 X 6 = 42

- 13. Describe the metabolism of polyamines; functions and clinical importance.
- 14. Describe on biosynthesis of Haem, add a note on regulation of Haem synthesis.
- 15. Describe the purine salvage pathway. Add a note Lesch Nyhan syndrome.
- 16. Mention about the principles, basic mechanism and tools of recombinant DNA technology.
- 17. Discuss on the classification of liver function tests in detail. And explain the mechanism of Vanden Berg test reaction.
- 18. Discuss how lifelong learning helps the growth of a physician.
- 19. Explain the mechanism of Hormone action.

### Very Short Answer Questions:

6 X 3 = 18

- 20. What is genetic code? Describe the characteristics of genetic code.
- 21. Detoxification by conjugation.
- 22. Structure of Immunoglobulin.
- 23. Describe the composition and functions of extracellular matrix.
- 24. Mention anticancer drugs used in chemotherapy and their mode of action.
- 25. Orotic Aciduria.

**KALOJI NARAYANA RAO UNIVERSITY OF HEALTH SCIENCES**  
TELANGANA STATE, WARANGAL – 506 002

FIRST MBBS DEGREE EXAMINATIONS: FEBRUARY, 2023

**BIOCHEMISTRY**  
**PAPER-I**

Max Marks: 100

Time: 3 Hours.

Note: Answer all questions.

Give Diagrammatic representation wherever necessary.

Write an essay on the following:

2 X 15 = 30

1. A 45-year-old man is brought to the hospital in a comatose state on examination. He was found to be dehydrated and toxic with characteristic breathing pattern and a sweet smell on his breath. He is admitted to the ward and investigated the following are the results of the lab investigations;

RBS- 380 mg %, Benedict's test with urine - positive brick red. Rothera's test with urine-positive. Plasma  $P^{H}$ - 7.25, Serum  $HCO_3^-$  - 15 meq/L

- Write the probable diagnosis?
- How do you interpret the above laboratory data?
- Discuss the pathophysiology of this case.
- What are the enzymes influenced by insulin. What are the derangements seen in diabetes mellitus.

(1+3+5+6)

2. a) Enumerate the major steps of synthesis of cholesterol.  
b) Name the rate limiting step of cholesterol synthesis.  
c) Explain how the cholesterol is transported from liver to peripheral tissues and back.  
d) Describe the role of cholesterol in atherosclerosis.

(7+1+2+5)

**Write a short note on the following:**

**8X5=40M**

- 3) Explain in detail the factors affecting the velocity of an enzyme action.
- 4) Humoral and cell mediated immunity.
- 5) Define detoxification. Give an account of various detoxification process.
- 6) Protein energy malnutrition.
- 7) Biochemical changes involving the metabolism of carbohydrates, fat and protein in starvation.
- 8) Describe the different types of ion channels and its significance.
- 9) Describe the structure and functions collagen. And give me two examples of diseases due to abnormalities of structure.
- 10) What are the biochemical functions and deficiency Manifestations of Vitamin D?

**Write briefly on the following**

**10X3=30M**

- 11) What is competitive inhibition? Explain with two examples.
- 12) What is redox potential and what is its significance.
- 13) What are the hormones influencing blood sugar levels and how are these hormones acting.
- 14) Compare and contrast three features of fatty acid synthesis and fatty acid oxidation.
- 15) Fluorosis.
- 16) Differentiate the biochemical properties of hexokinase and glucokinase.
- 17) What are the metabolic imbalances seen in diarrhoea.
- 18) Clinical application of therapeutic enzymes.
- 19) Classify Lipoproteins.
- 20) Define Oxidative phosphorylation. Explain uncouplers of oxidative phosphorylation.

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PAPER CODE: MB2019102

KALOJI NARAYANA RAO UNIVERSITY OF HEALTH SCIENCES  
TELANGANA STATE, WARANGAL – 506 002

FIRST MBBS DEGREE EXAMINATIONS: FEBRUARY, 2023

**BIOCHEMISTRY**  
**PAPER-II**

Time: 3 Hours.

Max Marks: 100

Note: Answer all questions.

Give Diagrammatic representation wherever necessary.

Write an essay on the following:

2 X 15 = 30

*Intermittent vomitings.*  
1. A 6-month-old infant began to vomit occasionally and ceased to gain weight. At 9-months of age he was readmitted to the hospital. Routine examination and laboratory analysis were normal. After one week, he became drowsy, had fever and elevated pulse. And there was hepatomegaly. EEG was done and was grossly abnormal. Blood ammonia was elevated and urine contained high amount of glutamine and uracil.

A) What is the probable diagnosis.

B) What is the mechanism underlying this disorder.

C) What is the normal ammonia level in the body.

D) Give details about ammonia detoxification.

E) Write about regulation of urea cycle.

(1+4+1+6+3)

2. What are the causes of hyperuricemia? Write the biochemical basis of clinical features of gout and its treatment modality. What are the normal values of serum uric acid. Add a note on hypouricemia.

(6+6+1+2)

**Write a short note on the following:**

**8 X 5 = 40**

3. Sickle cell disease.
4. Name the renal clearance tests. Give details of any one of them.
5. Phenylketonuria.
6. Write about the tumour markers. Add a note about telomeres.
7. Explain principle, procedure and applications of polymerase chain reaction (PCR).
8. Thyroid function tests.
9. Porphyrrias.
10. What are restriction enzymes? Explain by giving examples. What are their uses?

**Write briefly on the following**

**10 X 3 = 30**

11. Alkaptonuria.
12. Polyamines.
13. Anti Diuretic hormone
14. DNA mutations.
  - A. Mention the types of DNA mutations.
  - B. Justify the statement: "defects in DNA repair mechanisms leads to cancer".
15. Biologically active peptides.
16. Vandenberg Reaction.
17. Orotic Aciduria.
18. What is a Codon? Describe the salient features of genetic code.
19. Anion Gap.
20. Urinary Vanillyl Mandelic Acid (VMA) estimation is helpful for the diagnosis of Pheochromocytoma – Justify.

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KALOJI NARAYANA RAO UNIVERSITY OF HEALTH SCIENCES  
TELANGANA STATE, WARANGAL – 506 002

FIRST MBBS DEGREE SUPPELEMENTARY EXAMINATIONS: JUNE, 2023

**BIOCHEMISTRY**  
**PAPER-I**

Time: 3 Hours.

Max Marks: 100

Note: Answer all questions.

Give Diagrammatic representation wherever necessary.

Write an essay on the following:

2 X 15 = 30

1. A person is brought to the emergency department in a comatose state. There is history of prolonged fasting. The following test results were obtained.  
Blood glucose- 40 mgs/dl, Benedict's test - Negative, Rotheras test- Positive, Serum bicarbonate- 12 m.eq/L, Plasma  $P^H$  – 7.14
  - a) Write the probable diagnosis.
  - b) Describe the biochemical changes in carbohydrate and lipid metabolism in prolonged starvation.
  - c) Explain in detail the steps involved in gluconeogenesis and mention the key enzymes.
  - d) Regulation of gluconeogenesis and
  - e) Clinical significance of gluconeogenesis.

(1+4+4+3+3)
2.
  - a) Explain the components of electron transport chain.
  - b) Schematic representation of the flow of electrons through them.
  - c) Add a note on inhibitors and uncouplers.

(4+6+5)

**Write a short note on the following:**

**8 X 5 = 40**

3. Structure and functions of cholesterol and mention the products derived from cholesterol. Functions of cholesterol.
4. Discuss the renal mechanism involved in regulation of Blood pH.
5. Enumerate the mechanism of hormone action.
6. Electrophoresis and its clinical application.
7. Composition of extracellular matrix. Write about structure of collagen.
8. How will you perform glucose tolerance test (G.T.T) in a suspected case of diabetes mellitus. Describe the normal and abnormal G.T.T curves.
9. Classify the glycogen storage diseases. Elaborate the Vongierke's Disease.
10. Serum lipoproteins and its clinical importance.

**Write briefly on the following:**

**10 X 3 = 30**

11. Personal protective equipment and add a note on biosafety.
12. Name the second messengers. Draw the diagram of cyclic AMP.
13. Classify the phospholipids and their functions.
14. Thyroid hormones and their normal values.
15. Clinical significance of troponins.
16. Different transport mechanism across the cell membrane.
17. Classify the prostaglandins and their clinical application.
18. Role of glutathione in detoxification.
19. Define Immunity. Mention the types of immunoglobulins. Draw the structure of immunoglobulin.
20. Advantages and disadvantages of intake of polyunsaturated fatty acids.

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**BIOCHEMISTRY**  
**PAPER-II**

**Max Marks: 100**

**Time: 3 Hours.**

**Note: Answer all questions.**

**Give Diagrammatic representation wherever necessary.**

**2 X 15 = 30**

**Write an essay on the following:**

1. a. Describe the structure and functions of haemoglobin.  
b. Discuss about oxygen dissociation curve.  
c. Give an account on hemoglobinopathies with special reference to sickle-cell anemia. (4+3+8)
2. A 5-year-old fair, chubby boy was brought to the hospital with complaints of delayed developmental milestones, mental retardation, seizures and eczema. He exhibits light coloured hair, skin and eyes. Blood phenylalanine levels were more than 30 mg/dl (Normal 1-2 mg/dl). Guthrie bacterial inhibition assay and ferric chloride test showed blue green colour.
  - a) What is your probable diagnosis?
  - b) What is the cause of this disorder and treatment.
  - c) Give details about the metabolism of phenyl alanine?
  - d) Give an account on the biologically important substances produced from tyrosine?
  - e) Write on the inborn errors of their metabolism. (1+2+3+6+3)

8 X 5 = 40

Write a short essay on the following:

3. Give details about the formation of Uric acid and its clinical significance.
4. Formation of replication fork.
5. Write about inhibitors of protein synthesis.
6. Classify jaundice? Give an account of the biochemical tests which will help in differentiating types of jaundice.
7. Oncogenes.
8. Give an account of antioxidant minerals.
9. Functions and deficiency manifestations of Niacin.
10. Importance of creatinine clearance test.

10 X 3 = 30

Write briefly on the following:

11. Give an account on transmethylation reactions.
12. Characteristic features of genetic code.
13. Cell Cycle.
14. Vitamin K.
15. Factors regulating blood calcium level.
16. Write about differences between Marasmus and Kwashiorkor.
17. What is BMR? What are the factors affecting BMR.
18. Enumerate the thyroid function tests. Give details about TSH stimulation test.
19. DNA sequencing.
20. Importance of Dietary Fibre.

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FIRST MBBS DEGREE SUPPLEMENTARY EXAMINATIONS-JUNE, 2022  
**BIOCHEMISTRY (NEW REGULATION)**  
**PAPER – I**

Time: 3 Hours.

Max Marks: 100

Note: Answer all questions.

Give Diagrammatic representation wherever necessary.

Write a Long Answer Questions on the following

2 X 15 = 30

1. A Diabetic patient during a periodic check-up was found to have elevated plasma Cholesterol levels.

- What are the parameters measured under lipid profile?
- How is Cholesterol transported in plasma?
- Explain the clinical importance of controlling the plasma cholesterol level in a Diabetic patient.
- Name any two drugs to lower the plasma Cholesterol level.

(2+5+6+2)

2. Classify Enzymes with suitable examples. What are the factors affecting the enzyme action? Write briefly about enzyme inhibition.

(5+6+4)

**Medi circle**

**Write a Short Answer Questions on the following:**

**8X5=40**

3. Write about various components of electron transport chain and its inhibitors. ✓
4. Role of Lungs in maintenance of body pH. ✓
5. Mechanism of Group 1 hormone action. ✓
6. Write about the Regulation of Glycolysis. ✓
7. ELISA. ✓
8. Ketogenesis. ✓
9. Glycogen storage disorders. ✓
10. What is buffer and write about the various buffer systems present in the body. ✓

**Write Very Short Answer on the following:**

**10X3=30**

11. Give examples for Oxidative phosphorylation. ✓
12. Structure and Functions of Immunoglobulins. ✓
13. Define Epimers? Name two Epimers. ✓
14. Structure and Functions of Heparin. ✓
15. Detoxification by Conjugation. ✓
16. PDH Complex (Pyruvate dehydrogenase complex). ✓
17. Functions of Collagen. ✓
18. Essential Fatty acids. ✓
19. Galactosemia. ✓
20. Write about the different Transport Mechanisms present in the cell. ✓

**Medi circle**

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KALOJI NARAYANA RAO UNIVERSITY OF HEALTH SCIENCES  
WARANGAL, TELANGANA STATE – 506 002

FIRST MBBS DEGREE SUPPLEMENTARY EXAMINATIONS-JUNE, 2022  
**BIOCHEMISTRY (NEW REGULATION)**  
**PAPER – II**

Time: 3 Hours.

Max Marks: 100

Note: Answer all questions.

Give Diagrammatic representation wherever necessary.

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Write a Long Answer Questions on the following

2 X 15 = 30

1. Write in detail catabolism of Tyrosine. Write about the various products derived from Tyrosine. Add a note on disorders of Tyrosine catabolism. (5+5+5)
2. Discuss about the Vitamin A under the following. (2+2+1+5+5)
  - a) Chemistry.
  - b) Sources.
  - c) Daily requirement.
  - d) Biochemical functions.
  - e) Deficiency manifestations.

Medi circle

**Write a Short Answer Questions on the following:**

**8X5=40**

3. Regulation of Plasma calcium.
4. Purine salvage pathway.
5. Structure and functions of t-RNA.
6. Denaturation and Coagulation of Proteins.
7. Marasmus and Kwashiorkor.
8. Porphyrias.
9. Thyroid function tests.
10. Vanden Bergh reaction and its interpretation.

**Write Very Short Answer on the following:**

**10X3=30**

11. Biochemical functions of Zn and Iodine.
12. Haemoglobin Derivatives.
13. Genetic Code.
14. Write any 3 inhibitors of De Novo purine synthesis.
15. Phenyl Ketonuria.
16. Oncogenes.
17. Creatinine clearance.
18. Biochemical functions of Niacin coenzymes.
19. Homocystinuria.
20. Applications of PCR in Clinical Medicine.

**Medi circle**

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KALOJI NARAYANA RAO UNIVERSITY OF HEALTH SCIENCES  
WARANGAL, TELANGANA STATE – 506 002

FIRST MBBS DEGREE EXAMINATIONS-MARCH, 2022  
**BIOCHEMISTRY (NEW REGULATION)**  
**PAPER – I**

Time: 3 Hours.

Max Marks: 100

Note: Answer all questions.

Give Diagrammatic representation wherever necessary.

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Write a Long Answer Questions on the following

2 X 15 = 30

1. A 48 year old male presented to O.P with chest pain. Family history shows that his father died of a heart attack at the age of 46, and his elder brother also had a heart attack at the same age. The patient reports that he gets chest pain occasionally with ambulation and is not able to climb stairs without significant chest pain and shortness of breath. His plasma cholesterol level was 450 mgs%

- What is the possible diagnosis?
- What are the different types of lipoproteins and their biological significance.
- Discuss their role (LP) in cholesterol transport. How it is regulated.
- Normal values of HDL and LDL cholesterol.

(3+3+6+3)

2. a) State the blood buffer systems.

**Medi\_circle**

- What is clinical application of acid-base balance?
- Write in detail about renal and lung regulation of blood  $P^H$ .
- Classify acid base disturbances.
- What are the clinical conditions where you see metabolic acidosis?

(1+2+8+2+2)

**Write a Short Answer Questions on the following:**

8X5=40

3. Define active transport mechanism. Give 2 examples where drugs inhibit active transport.  
(1+2+2)
4. Formation of ketone bodies in our body and their utilization. How will you detect the ketone bodies in the urine?  
(2+2+1)
5. Describe Enzyme inhibition with examples.
6. Monoclonal antibodies.
7. Define hypoglycaemia clinical features, complications of hypoglycaemia. Normal levels of fasting blood sugar and Post prandial blood sugar and random blood sugar.  
(1+2+2)
8. Structure of collagen and elastin and their functions. Explain the biochemical basis of osteogenesis imperfecta.  
(2+1+2)
9. Explain HMP (Hexose Monophosphate Shunt ) Pathway. What is the significance of this pathway?  
(2+3)
10. Clinical importance of enzymes in assessment of cardiac disease & liver diseases. (2+3)

**Write Very Short Answer on the following:**

10X3=30

11. Biochemical markers of COVID 19.
12. Hormones derived from cholesterol.
13. Types of blotting techniques and their clinical application.
14. Name three Glycogen storage disorders.
15. Structure of proinsulin and insulin.
16. Normal values of serum electrolytes.
17. Rapoport-Leubering cycle and its importance.
18. Antioxidant Vitamins.
19. Detoxification by conjugation.
20. Essential fatty acids and their importance.  
(1+2)

**Medi\_circle**

**PAPER CODE: MB2019102**  
**KALOJI NARAYANA RAO UNIVERSITY OF HEALTH SCIENCES**  
**WARANGAL, TELANGANA STATE – 506 002**

**FIRST MBBS DEGREE EXAMINATIONS-MARCH, 2022**  
**BIOCHEMISTRY (NEW REGULATION)**  
**PAPER – II**

**Time: 3 Hours.**

**Max Marks: 100**

**Note: Answer all questions.**

**Give Diagrammatic representation wherever necessary.**

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**Write Long Answer on the following Questions:**

**2 X 15 = 30**

**1. Explain the process of Translation in Eucaryotes under following.**

**(3+3+3+3+3)**

- a) Activation of amino acid.
- b) Initiation.
- c) Elongation.
- d) Termination.
- e) Post translational Modifications.

**2. Discuss the IRON metabolism under the following.**

**(3+3+3+3+3)**

- a) Food sources.
- b) Daily requirement.
- c) Functions.
- d) Absorption and transport.
- e) Clinical abnormalities.

**Medi\_circle**

**Write a Short Answer on the following Questions:**

**8X5=40**

3. Heme synthesis.
4. Purine salvage pathway.
5. Polymerase chain reaction.
6. Transamination and its significance.
7. Polyamines.
8. Jaundice.
9. Urea cycle.
10. Tumour markers.

**Medi\_circle**

**Write Very Short Answer on the following Questions:**

**10X3=30**

11. Apoptosis.
12. Deficiency manifestations of Vitamin "D".
13. Clearance tests.
14. Write any three inhibitors of De Novo purine synthesis.
15. S-Adenosyl methionine.
16. Lesch-Nyhan syndrome and Orotic aciduria.
17. Write any three functions of ZINC.
18. Functions of Albumin.
19. Mention the biochemical defect in
  - a) Homocystinuria.
  - b) Phenylketonuria.
  - c) Maple syrup urine disease.
20. Functions of Vitamin 'C'.

\*\*\*\*\*

# M.B.B.S DEGREE EXAMINATION

July - 2019

First M.B.B.S Examination

BIOCHEMISTRY

PAPER - I

K  
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H  
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# M.B.B.S DEGREE EXAMINATION

July - 2019

First M.B.B.S Examination

BIOCHEMISTRY

PAPER - II

Time: 2½ Hours

Note : Answer all questions : Illustrate your answer with suitable diagrams.

## I. ANSWER THE FOLLOWING QUESTIONS

2 × 10 = 20

1. How ammonia is formed in the body ? Explain urea cycle. Add a note on ammonia toxicity. [3 + 4 + 3 = 10]

*Sanjivnarayana (5<sup>th</sup> Edn): 335 | Harper (30<sup>th</sup> Edn): 276-278 | DM Vasudevan (9<sup>th</sup> Edn): 272-275*

2. Explain how bile acids are synthesised ? Mention the functions of bile acids add a note on steatorrhea. [4 + 4 + 2 = 10]

*Sanjivnarayana (5<sup>th</sup> Edn): 454, 117 | Harper (30<sup>th</sup> Edn): 305 | DM Vasudevan (9<sup>th</sup> Edn): 367, 765*

## II. WRITE SHORT NOTE ON

[5 × 4 = 20]

3. Biochemical functions of glycine.

*Sanjivnarayana (5<sup>th</sup> Edn): 341 | DM Vasudevan (9<sup>th</sup> Edn): 29*

4. Amphibolic role of citric acid cycle

*Sanjivnarayana (5<sup>th</sup> Edn): 256, 237 | Harper (30<sup>th</sup> Edn): 161 | DM Vasudevan (9<sup>th</sup> Edn): 334*

5. Competitive inhibition of enzymes and its clinical importance

*Sanjivnarayana (5<sup>th</sup> Edn): 94 | Harper (30<sup>th</sup> Edn): 76-78 | DM Vasudevan (9<sup>th</sup> Edn): 65, 66*

6. Anti-oxidants

*Sanjivnarayana (5<sup>th</sup> Edn): 33, 132 | Harper (30<sup>th</sup> Edn): 532, 204 | DM Vasudevan (9<sup>th</sup> Edn): 476*

7. Hormonal regulation of blood glucose.

*Sanjivnarayana (5<sup>th</sup> Edn): 676 | Harper (30<sup>th</sup> Edn): 178-180 | DM Vasudevan (9<sup>th</sup> Edn): 135*

## III. WRITE BRIEFLY ON

[5 × 2 = 10]

8. Name the essential fatty acids

*Sanjivnarayana (5<sup>th</sup> Edn): 509 | DM Vasudevan (9<sup>th</sup> Edn): 255*

9. Secondary structure of proteins

*Sanjivnarayana (5<sup>th</sup> Edn): 56 | Harper (30<sup>th</sup> Edn): 34-41 | DM Vasudevan (9<sup>th</sup> Edn): 37*

10. Active transport system

*Sanjivnarayana (5<sup>th</sup> Edn): 627 | Harper (30<sup>th</sup> Edn): 468, 469 | DM Vasudevan (9<sup>th</sup> Edn): 21*

11. Sickle cell haemoglobin

*Sanjivnarayana (5<sup>th</sup> Edn): 203 | Harper (30<sup>th</sup> Edn): 397, 650, 54 | DM Vasudevan (9<sup>th</sup> Edn): 379*

12. Lactose intolerance

*Sanjivnarayana (5<sup>th</sup> Edn): 169 | Harper (30<sup>th</sup> Edn): 519-520 | DM Vasudevan (9<sup>th</sup> Edn): 128*

Max. Marks: 50

## I. ANSWER THE FOLLOWING QUESTIONS

2 × 10 = 20

1. Describe the process of protein biosynthesis in eukaryotes. Enumerate post translational modification. [7 + 3 = 10]

*Sanjivnarayana (5<sup>th</sup> Edn): 561 | Harper (30<sup>th</sup> Edn): 325 | DM Vasudevan (9<sup>th</sup> Edn): 619*

2. What are the sources of carbon and nitrogen atoms of pyrimidine ring? Describe biosynthesis of pyrimidine ribonucleotides. Add a note on its regulation

*Sanjivnarayana (5<sup>th</sup> Edn): 388 | Harper (30<sup>th</sup> Edn): 322 | DM Vasudevan (9<sup>th</sup> Edn): 576, 577, 583*

## II. WRITE SHORT NOTE ON [5 × 4 = 20]

3. Elaborate tests based on excretory function and synthetic Functions of liver

*Sanjivnarayana (5<sup>th</sup> Edn): 453, 459 | Harper (30<sup>th</sup> Edn): 566 | DM Vasudevan (9<sup>th</sup> Edn): 397*

4. Immunoglobulin's structure, types and functions

*Sanjivnarayana (5<sup>th</sup> Edn): 186 | Harper (30<sup>th</sup> Edn): 627 | DM Vasudevan (9<sup>th</sup> Edn): 693*

5. Protein energy malnutrition - features and biochemical investigations.

*Sanjivnarayana (5<sup>th</sup> Edn): 516 | DM Vasudevan (9<sup>th</sup> Edn): 555*

6. Describe the role of lungs and kidney in maintaining acid base balance

*Sanjivnarayana (5<sup>th</sup> Edn): 476, 477 | DM Vasudevan (9<sup>th</sup> Edn): 441-443*

7. What are oncogenes, proto - oncogenes and anti-oncogenes ? Name 2 oncofetal antigens and their significance.

*Sanjivnarayana (5<sup>th</sup> Edn): 659 | Harper (30<sup>th</sup> Edn): 683-688 | DM Vasudevan (9<sup>th</sup> Edn): 711*

## III. WRITE BRIEFLY ON

[5 × 2 = 10]

8. Acute intermittent porphyria - cause and feature

*Sanjivnarayana (5<sup>th</sup> Edn): 212 | Harper (30<sup>th</sup> Edn): 310 | DM Vasudevan (9<sup>th</sup> Edn): 364, 365*

9. Mutations types and effect

*Sanjivnarayana (5<sup>th</sup> Edn): 535 | DM Vasudevan (9<sup>th</sup> Edn): 630*

10. Tubular functions of Kidney

*Sanjivnarayana (5<sup>th</sup> Edn): 625 | Harper (30<sup>th</sup> Edn): 446 | DM Vasudevan (9<sup>th</sup> Edn): 648*

11. Gene therapy

*Sanjivnarayana (5<sup>th</sup> Edn): 599 | DM Vasudevan (9<sup>th</sup> Edn): 648*

*Sanjivnarayana (5<sup>th</sup> Edn): 625 | Harper (30<sup>th</sup> Edn): 446 | DM Vasudevan (9<sup>th</sup> Edn): 648*

*Sanjivnarayana (5<sup>th</sup> Edn): 599 | DM Vasudevan (9<sup>th</sup> Edn): 648*

Time: 2½ Hours

Note : Answer all questions : Illustrate your answer with suitable diagrams.

Max. Marks: 50

**I. ANSWER THE FOLLOWING QUESTIONS**

1. Enumerate source, daily requirement of "Vitamin-A". Describe functions and deficiency manifestations of Vitamin-A". [1 + 1 + 4 + 4 = 10]

Satyannarayana (5<sup>th</sup> Edn): 119, 122 | Harper (30<sup>th</sup> Edn): 529-530 | DM Vasudevan (9<sup>th</sup> Edn): 488-492

2. Define Jaundice. Mention types of jaundice, causes of jaundice, explain the findings in Urine and blood. [1 + 1 + 3 + 2 = 10]

Satyannarayana (5<sup>th</sup> Edn): 456 | Harper (30<sup>th</sup> Edn): 242 | DM Vasudevan (9<sup>th</sup> Edn): 370, 397**II. WRITE SHORT NOTE ON**

[5 × 4 = 20]

3. Chemiosmotic theory.

Satyannarayana (5<sup>th</sup> Edn): 229 | Harper (30<sup>th</sup> Edn): 121-124 | DM Vasudevan (9<sup>th</sup> Edn): 350-353

4. Define detoxification . Write in brief detoxification by different processes with examples.

Satyannarayana (5<sup>th</sup> Edn): 614 | Harper (30<sup>th</sup> Edn): 115-116 | DM Vasudevan (9<sup>th</sup> Edn): 479-482

5. Gluconeogenesis.

Satyannarayana (5<sup>th</sup> Edn): 258 | Harper (30<sup>th</sup> Edn): 172 | DM Vasudevan (9<sup>th</sup> Edn): 139

6. Schematic Presentation of Cholesterol Biosynthesis.

Satyannarayana (5<sup>th</sup> Edn): 309 | Harper (30<sup>th</sup> Edn): 249-253 | DM Vasudevan (9<sup>th</sup> Edn): 221

7. Enzyme Inhibition.

Satyannarayana (5<sup>th</sup> Edn): 87, 92 | Harper (30<sup>th</sup> Edn): 57, 76-79 | DM Vasudevan (9<sup>th</sup> Edn): 54, 65**III. WRITE BRIEFLY ON**

[5 × 2 = 10]

8. Enumerate four features of Marasmus.

Satyannarayana (5<sup>th</sup> Edn): 515 | Harper (30<sup>th</sup> Edn): 263, 524 | DM Vasudevan (9<sup>th</sup> Edn): 556

9. Conversion of Propionyl CoA to 4 Carbon compound

10. Biochemical markers for Myocardial Infarction.

Satyannarayana (5<sup>th</sup> Edn): 112 | Harper (30<sup>th</sup> Edn): 567 | DM Vasudevan (9<sup>th</sup> Edn): 240

11. Rappaport Lubering Cycle.

Satyannarayana (5<sup>th</sup> Edn): 251 | DM Vasudevan (9<sup>th</sup> Edn): 137

12. Hb Derivatives.

**I. ANSWER THE FOLLOWING QUESTIONS**

1. Describe the source, absorption, daily requirements, normal levels, distribution, functions and factors regulating blood calcium levels.

Satyannarayana (5<sup>th</sup> Edn): 406 | Harper (30<sup>th</sup> Edn): 506 | DM Vasudevan (9<sup>th</sup> Edn): 527-534

[1 + 1 + 1 + 1 + 3 + 3 = 10]

2. Structural organization of proteins.

Satyannarayana (5<sup>th</sup> Edn): 317 | Harper (30<sup>th</sup> Edn): 213, 239 | DM Vasudevan (9<sup>th</sup> Edn): 224

[2 + 4 + 3 + 1 = 10]

**II. WRITE SHORT NOTE ON**

[5 × 4 = 20]

3. Metabolic acids and respiratory alkalosis.

Satyannarayana (5<sup>th</sup> Edn): 481, 482 | Harper (30<sup>th</sup> Edn): 275 | DM Vasudevan (9<sup>th</sup> Edn): 446, 449

4. Tumor markers with examples.

Satyannarayana (5<sup>th</sup> Edn): 662 | Harper (30<sup>th</sup> Edn): 700-701 | DM Vasudevan (9<sup>th</sup> Edn): 715

5. Structure and function of immunoglobulin's.

Satyannarayana (5<sup>th</sup> Edn): 186 | Harper (30<sup>th</sup> Edn): 627 | DM Vasudevan (9<sup>th</sup> Edn): 693

6. Post transcriptional modification.

Satyannarayana (5<sup>th</sup> Edn): 547 | Harper (30<sup>th</sup> Edn): 338, 387 | DM Vasudevan (9<sup>th</sup> Edn): 611

7. Degradation of pyrimidines.

[5 × 2 = 10]

**III. WRITE BRIEFLY ON**

8. Difference between endonuclease and restriction endonuclease.

Harper (30<sup>th</sup> Edn): 361-369 | DM Vasudevan (9<sup>th</sup> Edn): 644

9. Source of carbon and nitrogen atoms in the purine ring.

Satyannarayana (5<sup>th</sup> Edn): 387 | DM Vasudevan (9<sup>th</sup> Edn): 575, 576

10. Nutritional classification of amino acid.

Satyannarayana (5<sup>th</sup> Edn): 44 | Harper (30<sup>th</sup> Edn): 282, 283 | DM Vasudevan (9<sup>th</sup> Edn): 28, 33

11. Xeroderma pigmentosa.

Satyannarayana (5<sup>th</sup> Edn): 538 | DM Vasudevan (9<sup>th</sup> Edn): 603

12. Features of genetic code.

Satyannarayana (5<sup>th</sup> Edn): 551 | Harper (30<sup>th</sup> Edn): 338 | DM Vasudevan (9<sup>th</sup> Edn): 615

**M.B.B.S DEGREE EXAMINATION**

August - 2018

First M.B.B.S Examination

**BIOCHEMISTRY****PAPER - I**

Time: 2½ Hours

Note: Answer all questions; Illustrate your answer with suitable diagrams.

1. Explain the formation of ATP in electron transport chain with diagram. Mention ATP synthesizing sites and inhibitors of ETC.

**[4 + 2 + 2 + 2 = 10]**Sanyanarayana (5<sup>th</sup> Edn): 224-225, 227, 233 | Harper (30<sup>th</sup> Edn): 126-133 | Harper (31<sup>st</sup> Edn): 118-126  
DM Vasudevan (9<sup>th</sup> Edn): 346-354

2. Describe the degradation of glycogen in the body with regulation. Add a note on glycogen storage disorders.

**[4 + 2 + 4 = 10]**Sanyanarayana (5<sup>th</sup> Edn): 263, 269 | Harper (30<sup>th</sup> Edn): 178, 183 | Harper (31<sup>st</sup> Edn): 164-171 |  
DM Vasudevan (9<sup>th</sup> Edn): 143, 147**WRITE SHORT NOTE ON**

3. Hemoglobinopathies

**[5 × 4 = 20]**

4. Ketone bodies as fuel

Sanyanarayana (5<sup>th</sup> Edn): 203 | Harper (30<sup>th</sup> Edn): 57, 694 | Harper (31<sup>st</sup> Edn): 651 | DM Vasudevan (9<sup>th</sup> Edn): 378

5. Thiamine sources, functions and deficiency manifestation.

Sanyanarayana (5<sup>th</sup> Edn): 138 | Harper (30<sup>th</sup> Edn): 555, 556 | Harper (31<sup>st</sup> Edn): 533, 534 |  
DM Vasudevan (9<sup>th</sup> Edn): 502-503

6. Isoenzymes

Sanyanarayana (5<sup>th</sup> Edn): 112 | Harper (30<sup>th</sup> Edn): 66 | Harper (31<sup>st</sup> Edn): 56-66 | DM Vasudevan (9<sup>th</sup> Edn): 72

7. PEM and Balanced diet.

Sanyanarayana (5<sup>th</sup> Edn): 514 | Harper (30<sup>th</sup> Edn): 421 | DM Vasudevan (9<sup>th</sup> Edn): 555**WRITE BRIEFLY ON**

8. Two reactions of detoxification by Hydrolysis.

**[5 × 2 = 10]**Sanyanarayana (5<sup>th</sup> Edn): 615 | Harper (30<sup>th</sup> Edn): 586 | Harper (31<sup>st</sup> Edn): 556 | DM Vasudevan (9<sup>th</sup> Edn): 480

9. Two functional roles of Vitamin B<sub>12</sub>

Sanyanarayana (5<sup>th</sup> Edn): 156 | Harper (30<sup>th</sup> Edn): 558 | Harper (31<sup>st</sup> Edn): 528 | DM Vasudevan (9<sup>th</sup> Edn): 516

10. Flip ratio

11. Brown fat

Sanyanarayana (5<sup>th</sup> Edn): 231, 517

12. Functions of Hyaluronic acid.

Sanyanarayana (5<sup>th</sup> Edn): 23 | Harper (30<sup>th</sup> Edn): 156 | Harper (31<sup>st</sup> Edn): 599 | DM Vasudevan (9<sup>th</sup> Edn): 95, 96K  
N  
R  
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S**M.B.B.S DEGREE EXAMINATION**

August - 2018

First M.B.B.S Examination

**BIOCHEMISTRY****PAPER - II**

Note: Answer all questions; Illustrate your answer with suitable diagrams.

1. Describe the biological products formed from Tyrosine and tryptophan.

Max. Marks: 50

**[10]**Sanyanarayana (5<sup>th</sup> Edn): 346, 355 | Harper (30<sup>th</sup> Edn): 304, 316, 317 | Harper (31<sup>st</sup> Edn): 286 |  
DM Vasudevan (9<sup>th</sup> Edn): 306

2. Describe the mucosal therapy of iron absorption.

**[10]**Sanyanarayana (5<sup>th</sup> Edn): 416 | Harper (30<sup>th</sup> Edn): 541 | DM Vasudevan (9<sup>th</sup> Edn): 537**WRITE SHORT NOTE ON**

3. Types of DNA

**[5 × 4 = 20]**

4. Specialized products derived from glycine.

Sanyanarayana (5<sup>th</sup> Edn): 76 | Harper (30<sup>th</sup> Edn): 359-363 | Harper (31<sup>st</sup> Edn): 338 | DM Vasudevan (9<sup>th</sup> Edn): 593

5. Secondary active transport.

Sanyanarayana (5<sup>th</sup> Edn): 342 | Harper (30<sup>th</sup> Edn): 283 | Harper (31<sup>st</sup> Edn): 282 | DM Vasudevan (9<sup>th</sup> Edn): 283

6. Polymerization chain reaction.

Sanyanarayana (5<sup>th</sup> Edn): 626

7. Western blotting.

Sanyanarayana (5<sup>th</sup> Edn): 534 | Harper (30<sup>th</sup> Edn): 70, 458, 459 | Harper (31<sup>st</sup> Edn): 439 |  
DM Vasudevan (9<sup>th</sup> Edn): 658

8. Functions of Nucleotides.

Sanyanarayana (5<sup>th</sup> Edn): 577 | Harper (31<sup>st</sup> Edn): 438**WRITE BRIEFLY ON**

9. Protein folding.

Sanyanarayana (5<sup>th</sup> Edn): 69 | Harper (30<sup>th</sup> Edn): 339-346 | Harper (31<sup>st</sup> Edn): 321 |  
DM Vasudevan (9<sup>th</sup> Edn): 576, 577

10. Obstructive Jaundice.

Sanyanarayana (5<sup>th</sup> Edn): 556

11. Isoenzymes

Sanyanarayana (5<sup>th</sup> Edn): 456 | Harper (30<sup>th</sup> Edn): 323 | Harper (31<sup>st</sup> Edn): 242 | DM Vasudevan (9<sup>th</sup> Edn): 571

12. Liposomes

Sanyanarayana (5<sup>th</sup> Edn): 112 | Harper (30<sup>th</sup> Edn): 66 | DM Vasudevan (9<sup>th</sup> Edn): 72

**M.B.B.S DEGREE EXAMINATION**

July - 2018

First M.B.B.S Examination

**BIOCHEMISTRY****PAPER - I****M.B.B.S DEGREE EXAMINATION**

July - 2018

First M.B.B.S Examination

**BIOCHEMISTRY****PAPER - II****Time: 2½ Hours****Note: Answer all questions: Illustrate your answer with suitable diagrams.****Max. Marks: 50**

1. Explain the classification of enzymes citing examples. Give an account on the enzymes of diagnostic importance. [5 + 5 = 10]

Sayanarayana (5<sup>th</sup> Edn): 88, 89 | Harper (31<sup>st</sup> Edn): 57, 64 | DM Vasudevan (9<sup>th</sup> Edn): 33, 34

2. Describe the metabolism of low density lipoprotein (LDL). Add a note on hypercholesterolemia. [7 + 3 = 10]

Sayanarayana (5<sup>th</sup> Edn): 39, 319, 317 | Harper (31<sup>st</sup> Edn): 237, 257**WRITE SHORT NOTE ON**

[5 × 4 = 20]

3. Homopolysaccharides

Sayanarayana (5<sup>th</sup> Edn): 10, 21 | Harper (31<sup>st</sup> Edn): 142 | DM Vasudevan (9<sup>th</sup> Edn): 93, 94

4. Electron transport chain

Sayanarayana (5<sup>th</sup> Edn): 223 | Harper (31<sup>st</sup> Edn): 118 | DM Vasudevan (9<sup>th</sup> Edn): 346-354

5. Functions of vitamin C

Sayanarayana (5<sup>th</sup> Edn): 135-137 | Harper (31<sup>st</sup> Edn): 528, 539 | DM Vasudevan (9<sup>th</sup> Edn): 519-521

6. Importance of HMP pathway

Sayanarayana (5<sup>th</sup> Edn): 272 | Harper (31<sup>st</sup> Edn): 182 | DM Vasudevan (9<sup>th</sup> Edn): 147

7. Bilirubin excretion

Sayanarayana (5<sup>th</sup> Edn): 455 | Harper (31<sup>st</sup> Edn): 312 | DM Vasudevan (9<sup>th</sup> Edn): 369**WRITE BRIEFLY ON**

[5 × 2 = 10]

8. List any four pathways located in mitochondria

9. Functions of vitamin K

Sayanarayana (5<sup>th</sup> Edn): 134 | Harper (31<sup>st</sup> Edn): 528, 532-533 | DM Vasudevan (9<sup>th</sup> Edn): 478

10. Importance of Rapoport Luebering cycle

Sayanarayana (5<sup>th</sup> Edn): 199 | DM Vasudevan (9<sup>th</sup> Edn): 137

11. List any four causes of hyperbilirubinemia

Harper (31<sup>st</sup> Edn): 305, 313-314

12. List any four conjugating agents of phase II detoxification

Sayanarayana (5<sup>th</sup> Edn): 615 | Harper (31<sup>st</sup> Edn): 528 | DM Vasudevan (9<sup>th</sup> Edn): 429

1. Describe the replication of DNA with illustration. [3 + 2 = 10]

Sayanarayana (5<sup>th</sup> Edn): 524-528 | Harper (31<sup>st</sup> Edn): 361 | DM Vasudevan (9<sup>th</sup> Edn): 594-606

2. Discuss the role of buffers and kidney in pH homeostasis. [7 + 3 = 10]

Sayanarayana (5<sup>th</sup> Edn): 647**WRITE SHORT NOTE ON**

[5 × 4 = 20]

3. Liver function tests

Sayanarayana (5<sup>th</sup> Edn): 454 | Harper (31<sup>st</sup> Edn): 566 | DM Vasudevan (9<sup>th</sup> Edn): 597

4. Metabolic acidosis

Sayanarayana (5<sup>th</sup> Edn): 493 | Harper (31<sup>st</sup> Edn): 575 | DM Vasudevan (9<sup>th</sup> Edn): 446

5. Purine catabolism

Sayanarayana (5<sup>th</sup> Edn): 395 | Harper (31<sup>st</sup> Edn): 334 | DM Vasudevan (9<sup>th</sup> Edn): 582

6. Structure of immunoglobulin

Sayanarayana (5<sup>th</sup> Edn): 187 | Harper (31<sup>st</sup> Edn): 639 | DM Vasudevan (9<sup>th</sup> Edn): 693

7. Structure of plasma membrane

Sayanarayana (5<sup>th</sup> Edn): 623 | Harper (31<sup>st</sup> Edn): 459 | DM Vasudevan (9<sup>th</sup> Edn): 15**WRITE BRIEFLY ON**

[5 × 2 = 10]

8. List the special products derived from glycine

Sayanarayana (5<sup>th</sup> Edn): 344 | Harper (31<sup>st</sup> Edn): 534-551 | DM Vasudevan (9<sup>th</sup> Edn): 283

9. Phenylketonuria

Sayanarayana (5<sup>th</sup> Edn): 352 | Harper (31<sup>st</sup> Edn): 204, 208 | DM Vasudevan (9<sup>th</sup> Edn): 249

10. Write the normal values of serum sodium and potassium

Sayanarayana (5<sup>th</sup> Edn): 413, 414 | Harper (31<sup>st</sup> Edn): 492 | DM Vasudevan (9<sup>th</sup> Edn): 463, 462

11. List four iron containing proteins

12. List four antioxidants

Sayanarayana (5<sup>th</sup> Edn): 632 | Harper (31<sup>st</sup> Edn): 315 | DM Vasudevan (9<sup>th</sup> Edn): 475

# M.B.B.S DEGREE EXAMINATION

November/December - 2017

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - I

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# M.B.B.S DEGREE EXAMINATION

November/December - 2017

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - II

Time: 2½ Hours

Note : Answer all questions : Illustrate your answer with suitable diagrams.

Max. Marks: 50

1. Describe HMP pathway, its metabolic significance and add a note on glucose6 phosphate dehydrogenase deficiency. [6+2+2=10]

Satyanarayana (5<sup>th</sup> Edn): 270, 274 | Harper (31<sup>st</sup> Edn): 180-185, 335 | DM Vasudevan (9<sup>th</sup> Edn): 147

2. Classify enzymes with examples and write in detail the diagnostic importance of enzymes. [6+4=10]

Satyanarayana (5<sup>th</sup> Edn): 88, 92 | Harper (31<sup>st</sup> Edn): 51, 63-65 | DM Vasudevan (9<sup>th</sup> Edn): 53, 54

### WRITE SHORT NOTE ON

[5 x 4 = 20]

3. Functions of cholesterol

Satyanarayana (5<sup>th</sup> Edn): 309 | Harper (31<sup>st</sup> Edn): 202 | DM Vasudevan (9<sup>th</sup> Edn): 221

4. Pellagra

Satyanarayana (5<sup>th</sup> Edn): 141 | Harper (31<sup>st</sup> Edn): 534, 535 | DM Vasudevan (9<sup>th</sup> Edn): 507

5. Hemoglobinopathies

Satyanarayana (5<sup>th</sup> Edn): 203 | Harper (31<sup>st</sup> Edn): 53, 54 | DM Vasudevan (9<sup>th</sup> Edn): 378

6. Oxidative phosphorylation

Satyanarayana (5<sup>th</sup> Edn): 228 | Harper (31<sup>st</sup> Edn): 620 | DM Vasudevan (9<sup>th</sup> Edn): 350

7. Kwashiorkor

Satyanarayana (5<sup>th</sup> Edn): 515 | Harper (31<sup>st</sup> Edn): 263, 264, 524, 525 | DM Vasudevan (9<sup>th</sup> Edn): 555, 556

### WRITE BRIEFLY ON

[5 x 2 = 10]

8. Lipotropic factors

Satyanarayana (5<sup>th</sup> Edn): 324 | Harper (31<sup>st</sup> Edn): 244 | DM Vasudevan (9<sup>th</sup> Edn): 212

9. Name the bile salts and bile pigments

Satyanarayana (5<sup>th</sup> Edn): 174, 214 | Harper (31<sup>st</sup> Edn): 521, 305 | DM Vasudevan (9<sup>th</sup> Edn): 231

10. Name four functions of vitamin C

Satyanarayana (5<sup>th</sup> Edn): 135 | Harper (31<sup>st</sup> Edn): 539 | DM Vasudevan (9<sup>th</sup> Edn): 519, 520, 521

11. Steatorrhea

Satyanarayana (5<sup>th</sup> Edn): 117 | DM Vasudevan (9<sup>th</sup> Edn): 765

12. Specific dynamic action

Satyanarayana (5<sup>th</sup> Edn): 505 | Harper (31<sup>st</sup> Edn): 156, 520, 145 | DM Vasudevan (9<sup>th</sup> Edn): 551

1. Describe in detailed the metabolism of phenylalanine and add a note on phenyl ketonuria. [7 + 3 = 10]

Satyanarayana (5<sup>th</sup> Edn): 345 | Harper (31<sup>st</sup> Edn): 284 | DM Vasudevan (9<sup>th</sup> Edn): 316-319, 320

2. Give an account of daily requirement, absorption, transport, biochemical functions and clinical abnormalities of iron. [1 + 2 + 2 + 3 = 8]

Satyanarayana (5<sup>th</sup> Edn): 415 | Harper (31<sup>st</sup> Edn): 524, 571, 581 | DM Vasudevan (9<sup>th</sup> Edn): 536-545

### WRITE SHORT NOTE ON

[5 x 4 = 20]

3. Types of RNA and explain functions of each

Satyanarayana (5<sup>th</sup> Edn): 31 | Harper (31<sup>st</sup> Edn): 386 | DM Vasudevan (9<sup>th</sup> Edn): 508-513

4. Quaternary Structure of protein

Satyanarayana (5<sup>th</sup> Edn): 55, 61 | Harper (31<sup>st</sup> Edn): 34 | DM Vasudevan (9<sup>th</sup> Edn): 35

5. Metabolism of Glycine

Satyanarayana (5<sup>th</sup> Edn): 343 | Harper (31<sup>st</sup> Edn): 302-307 | DM Vasudevan (9<sup>th</sup> Edn): 282, 283

6. Fluid mosaic model of membrane structure

Satyanarayana (5<sup>th</sup> Edn): 623 | DM Vasudevan (9<sup>th</sup> Edn): 15

7. Recombinant DNA and its applications.

Satyanarayana (5<sup>th</sup> Edn): 578 | Harper (31<sup>st</sup> Edn): 433 | DM Vasudevan (9<sup>th</sup> Edn): 543-547

### WRITE BRIEFLY ON

[5 x 2 = 10]

8. Bence Jones Protein

Satyanarayana (5<sup>th</sup> Edn): 189

9. Differences between Nucleotide and Nucleoside

Satyanarayana (5<sup>th</sup> Edn): 72, 73 | Harper (31<sup>st</sup> Edn): 332, 320, 349

10. Zymogens

Satyanarayana (5<sup>th</sup> Edn): 562 | Harper (31<sup>st</sup> Edn): 87, 927 | DM Vasudevan (9<sup>th</sup> Edn): 281

11. Vandenberg reaction

Satyanarayana (5<sup>th</sup> Edn): 456 | DM Vasudevan (9<sup>th</sup> Edn): 398

12. Active Transport

Satyanarayana (5<sup>th</sup> Edn): 627 | Harper (31<sup>st</sup> Edn): 467 | DM Vasudevan (9<sup>th</sup> Edn): 21

**M.B.B.S DEGREE EXAMINATION**

August - 2017

**First M.B.B.S Examination**

**BIOCHEMISTRY**

**PAPER - I**

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**M.B.B.S DEGREE EXAMINATION**

August - 2017

**First M.B.B.S Examination**

**BIOCHEMISTRY**

**PAPER - II**

**Time: 2½ Hours**

**Note : Answer all questions : Illustrate your answer with suitable diagrams.**

**Max. Marks: 50**

1. Describe the tricarboxylic acid cycle and explain its significance. Add a note on the energetics. [5+5=10]

Satyanarayana (5<sup>th</sup> Edn): 254, 256 | Harper (31<sup>st</sup> Edn): 130 | DM Vasudevan (9<sup>th</sup> Edn): 334.

2. Describe the sources, dietary requirements, biochemical functions and deficiency manifestations of Vitamin A. [1+2+5+2=10]

Satyanarayana (5<sup>th</sup> Edn): 119 | Harper (31<sup>st</sup> Edn): 529-530 | DM Vasudevan (9<sup>th</sup> Edn): 488, 492

**WRITE SHORT NOTE ON**

[5 × 4 = 20]

3. Iso enzymes and their clinical importance:

Satyanarayana (5<sup>th</sup> Edn): 112 | Harper (31<sup>st</sup> Edn): 61-62 | DM Vasudevan (9<sup>th</sup> Edn): 72

4. Porphyrins

Satyanarayana (5<sup>th</sup> Edn): 210 | Harper (31<sup>st</sup> Edn): 305-310 | DM Vasudevan (9<sup>th</sup> Edn): 363

5. Dietary fibre

Satyanarayana (5<sup>th</sup> Edn): 508 | DM Vasudevan (9<sup>th</sup> Edn): 552

6. High energy compounds

Satyanarayana (5<sup>th</sup> Edn): 220 | DM Vasudevan (9<sup>th</sup> Edn): 345

7. Detoxification by conjugation

Satyanarayana (5<sup>th</sup> Edn): 614 | Harper (31<sup>st</sup> Edn): 5558 | DM Vasudevan (9<sup>th</sup> Edn): 480

**WRITE BRIEFLY ON**

[5 × 2 = 10]

8. Essential fatty acids

Satyanarayana (5<sup>th</sup> Edn): 509 | Harper (31<sup>st</sup> Edn): 196, 216 | DM Vasudevan (9<sup>th</sup> Edn): 255

9. Causes of fatty liver

Satyanarayana (5<sup>th</sup> Edn): 322 | Harper (31<sup>st</sup> Edn): 243 | DM Vasudevan (9<sup>th</sup> Edn): 211, 212

10. Coenzyme form of Niacin, write one reaction.

Satyanarayana (5<sup>th</sup> Edn): 142 | Harper (31<sup>st</sup> Edn): 528 | DM Vasudevan (9<sup>th</sup> Edn): 502, 503

11. Lactose intolerance.

Satyanarayana (5<sup>th</sup> Edn): 171 | Harper (31<sup>st</sup> Edn): 519-520 | DM Vasudevan (9<sup>th</sup> Edn): 128

12. What are Ketone bodies and write about Ketonuria.

Satyanarayana (5<sup>th</sup> Edn): 294 | Harper (31<sup>st</sup> Edn): 215, 290, 207, 210 | DM Vasudevan (9<sup>th</sup> Edn): 213

1. Enumerate the liver function tests and kidney function tests. Write in detail about one kidney function test and one liver function test.

[4 + 3 + 3 = 10]

Satyanarayana (5<sup>th</sup> Edn): 453-459 | Harper (31<sup>st</sup> Edn): 566 | DM Vasudevan (9<sup>th</sup> Edn): 397

2. Write in detail about urea cycle and add a note on urea cycle disorders.

[6 + 4 = 10]

Satyanarayana (5<sup>th</sup> Edn): 336, 337 | Harper (31<sup>st</sup> Edn): 272 | DM Vasudevan (9<sup>th</sup> Edn): 272-275

**WRITE SHORT NOTE ON**

[5 × 4 = 20]

3. Polymerase chain reaction (PCR)

Satyanarayana (5<sup>th</sup> Edn): 534 | Harper (31<sup>st</sup> Edn): 439, 440 | DM Vasudevan (9<sup>th</sup> Edn): 658

4. Gout

Satyanarayana (5<sup>th</sup> Edn): 395 | Harper (31<sup>st</sup> Edn): 190, 224, 334 | DM Vasudevan (9<sup>th</sup> Edn): 583

5. Metabolic acidosis

Satyanarayana (5<sup>th</sup> Edn): 481 | Harper (31<sup>st</sup> Edn): 275 | DM Vasudevan (9<sup>th</sup> Edn): 446

6. Differences between DNA and RNA

Satyanarayana (5<sup>th</sup> Edn): 73 | Harper (31<sup>st</sup> Edn): 370, 362

7. Calcitriol

Satyanarayana (5<sup>th</sup> Edn): 409 | Harper (31<sup>st</sup> Edn): 530-531

**WRITE BRIEFLY ON**

[5 × 2 = 10]

8. Wilson's disease

Satyanarayana (5<sup>th</sup> Edn): 417 | Harper (31<sup>st</sup> Edn): 627 | DM Vasudevan (9<sup>th</sup> Edn): 430, 541

9. Orotic aciduria

Satyanarayana (5<sup>th</sup> Edn): 400 | Harper (31<sup>st</sup> Edn): 336 | DM Vasudevan (9<sup>th</sup> Edn): 587

10. Four factors effecting absorption of iron.

Satyanarayana (5<sup>th</sup> Edn): 416 | Harper (31<sup>st</sup> Edn): 523, 524 | DM Vasudevan (9<sup>th</sup> Edn): 537

11. Anion gap

Satyanarayana (5<sup>th</sup> Edn): 480 | DM Vasudevan (9<sup>th</sup> Edn): 176, 446

12. Flourosis.

Satyanarayana (5<sup>th</sup> Edn): 420 | DM Vasudevan (9<sup>th</sup> Edn): 542, 543

# M.B.B.S DEGREE EXAMINATION

December - 2016

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - I

# M.B.B.S DEGREE EXAMINATION

December - 2016

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - II

Time: 2½ Hours

Note : Answer all questions : Illustrate your answer with suitable diagrams.

Max. Marks: 50

1. Write in detail about the metabolism of chylomicrons giving suitable examples. [10]

Satyannarayana (5<sup>th</sup> Edn): 177, 319 | Harper (31<sup>st</sup> Edn): 240-241

2. Write in detail about the steps of glycolysis in anaerobic condition. Add a note on its regulation and energetics. [10]

Satyannarayana (5<sup>th</sup> Edn): 244, 353 | Harper (31<sup>st</sup> Edn): 157-163

3. Functions of biotin [5 × 4 = 20]

Satyannarayana (5<sup>th</sup> Edn): 150 | Harper (31<sup>st</sup> Edn): 528, 538 | DM Vasudevan (9<sup>th</sup> Edn): 511

4. Homopolysaccharides

Satyannarayana (5<sup>th</sup> Edn): 10, 21 | Harper (31<sup>st</sup> Edn): 156-159 | DM Vasudevan (9<sup>th</sup> Edn): 93, 94

5. Define Basal Metabolic Rate (BMR) and write the factors affecting BMR

Satyannarayana (5<sup>th</sup> Edn): 502 | Harper (31<sup>st</sup> Edn): 524 | DM Vasudevan (9<sup>th</sup> Edn): 550

6. Protein energy malnutrition

Satyannarayana (5<sup>th</sup> Edn): 515 | DM Vasudevan (9<sup>th</sup> Edn): 555

7. Isoenzymes

Satyannarayana (5<sup>th</sup> Edn): 112, 114 | Harper (31<sup>st</sup> Edn): 61-62 | DM Vasudevan (9<sup>th</sup> Edn): 72

8. Significance of uronic acid pathway [5 × 2 = 10]

Satyannarayana (5<sup>th</sup> Edn): 275 | Harper (31<sup>st</sup> Edn): 182, 189

9. Acute intermittent porphyria-mention the deficient enzyme and the lab findings

Satyannarayana (5<sup>th</sup> Edn): 210 | Harper (31<sup>st</sup> Edn): 310 | DM Vasudevan (9<sup>th</sup> Edn): 364

10. Effect of temperature on enzyme activity

Satyannarayana (5<sup>th</sup> Edn): 91 | Harper (31<sup>st</sup> Edn): 75 | DM Vasudevan (9<sup>th</sup> Edn): 65

11. Mention different types of α-Thalassemias

Satyannarayana (5<sup>th</sup> Edn): 205 | Harper (31<sup>st</sup> Edn): 54, 650 | DM Vasudevan (9<sup>th</sup> Edn): 381

12. Mention functions of Haemoglobin

Satyannarayana (5<sup>th</sup> Edn): 197, 206 | Harper (31<sup>st</sup> Edn): 53

1. Write about transamination deamination and transmethylation [4 + 3 + 3 = 10]

Satyannarayana (5<sup>th</sup> Edn): 333, 361, 334 | Harper (31<sup>st</sup> Edn): 131 | DM Vasudevan (9<sup>th</sup> Edn): 270, 271

2. Discussion the role of buffers and kidney in pH homeostasis [3 + 7 = 10]

Satyannarayana (5<sup>th</sup> Edn): 476, 678

- WRITE SHORT NOTES ON [5 × 4 = 20]

- (a) Hyperuricemia

Satyannarayana (5<sup>th</sup> Edn): 269 | Harper (31<sup>st</sup> Edn): 334

- (b) Mutations

Satyannarayana (5<sup>th</sup> Edn): 534 | Harper (31<sup>st</sup> Edn): 57-58 | DM Vasudevan (9<sup>th</sup> Edn): 630

- (c) Structure of plasma membrane

Satyannarayana (5<sup>th</sup> Edn): 623 | Harper (31<sup>st</sup> Edn): 459 | DM Vasudevan (9<sup>th</sup> Edn): 15

- (d) Recombinant DNA

Satyannarayana (5<sup>th</sup> Edn): 571 | Harper (31<sup>st</sup> Edn): 441 | DM Vasudevan (9<sup>th</sup> Edn): 643

- (e) Iron absorption

Satyannarayana (5<sup>th</sup> Edn): 136, 416 | Harper (31<sup>st</sup> Edn): 523 | DM Vasudevan (9<sup>th</sup> Edn): 556

- WRITE BRIEFLY ON [5 × 2 = 10]

- (a) Isoelectric Ph

Satyannarayana (5<sup>th</sup> Edn): 52, 64 | Harper (31<sup>st</sup> Edn): 19 | DM Vasudevan (9<sup>th</sup> Edn): 42

- (b) Functions of Calcium

Satyannarayana (5<sup>th</sup> Edn): 128 | Harper (31<sup>st</sup> Edn): 523 | DM Vasudevan (9<sup>th</sup> Edn): 527-534

- (c) Termination of Transcription

Satyannarayana (5<sup>th</sup> Edn): 541 | Harper (31<sup>st</sup> Edn): 383

- (d) Alkaptonuria

Satyannarayana (5<sup>th</sup> Edn): 353 | Harper (31<sup>st</sup> Edn): 288 | DM Vasudevan (9<sup>th</sup> Edn): 511

- (e) Mention functions of immunoglobulins

Satyannarayana (5<sup>th</sup> Edn): 157, 575 | Harper (31<sup>st</sup> Edn): 627 | DM Vasudevan (9<sup>th</sup> Edn): 485, 490

**M.B.B.S DEGREE EXAMINATION**

August - 2016

**First M.B.B.S Examination****BIOCHEMISTRY****PAPER - I****M.B.B.S DEGREE EXAMINATION**

August - 2016

**First M.B.B.S Examination****BIOCHEMISTRY****PAPER - II****Max. Marks: 50****Time: 2½ Hours****Note : Answer all questions : Illustrate your answer with suitable diagrams.**

1. Write the classification of enzymes giving examples. Give an account on enzyme inhibition? [5+5=10]

Satyanarayana (5<sup>th</sup> Edn): 87-92 | Harper (31<sup>st</sup> Edn): 63-65 | DM Vasudevan (9<sup>th</sup> Edn): 53, 54

2. Write the definition, substrates, pathway including enzymes and regulation of gluconeogenesis? [1+2+5+2=10]

Satyanarayana (5<sup>th</sup> Edn): 258 | Harper (31<sup>st</sup> Edn): 130, 133, 161 | DM Vasudevan (9<sup>th</sup> Edn): 139**WRITE SHORT NOTE ON****[5 × 4 = 20]**

3. Chemiosmotic hypothesis.

Satyanarayana (5<sup>th</sup> Edn): 228 | Harper (31<sup>st</sup> Edn): 121 | DM Vasudevan (9<sup>th</sup> Edn): 350-351

4. Phase I of detoxification.

Satyanarayana (5<sup>th</sup> Edn): 276, 344, 415, 613 | Harper (31<sup>st</sup> Edn): 115, 116 | DM Vasudevan (9<sup>th</sup> Edn): 479

5. Bile acids

Satyanarayana (5<sup>th</sup> Edn): 174, 315 | Harper (31<sup>st</sup> Edn): 202, 249 | DM Vasudevan (9<sup>th</sup> Edn): 367

6. Sickle cell anemia

Satyanarayana (5<sup>th</sup> Edn): 201, 608 | Harper (31<sup>st</sup> Edn): 650 | DM Vasudevan (9<sup>th</sup> Edn): 379

7. Dietary fibre - beneficials and adverse effects.

Satyanarayana (5<sup>th</sup> Edn): 170, 318, 507 | Harper (31<sup>st</sup> Edn): 526 | DM Vasudevan (9<sup>th</sup> Edn): 552**WRITE BRIEFLY ON****[5 × 2 = 10]**

8. Functions of vitamin K

Satyanarayana (5<sup>th</sup> Edn): 132 | Harper (31<sup>st</sup> Edn): 532-533 | DM Vasudevan (9<sup>th</sup> Edn): 498

9. Steatorrhea

Satyanarayana (5<sup>th</sup> Edn): 177 | DM Vasudevan (9<sup>th</sup> Edn): 765

10. Von-Gierkes disease

Satyanarayana (5<sup>th</sup> Edn): 269 | Harper (31<sup>st</sup> Edn): 335 | DM Vasudevan (9<sup>th</sup> Edn): 147

11. Deficiency manifestation of folic acid

Satyanarayana (5<sup>th</sup> Edn): 153 | Harper (31<sup>st</sup> Edn): 282 | DM Vasudevan (9<sup>th</sup> Edn): 512

12. Lipoprotein structure.

Satyanarayana (5<sup>th</sup> Edn): 319 | Harper (31<sup>st</sup> Edn): 237-238 | DM Vasudevan (9<sup>th</sup> Edn): 224

1. Write in detail about the metabolism of phenylalanine. Add a note on phenylketonuria? [7 + 3 = 10]

Satyanarayana (5<sup>th</sup> Edn): 347, 352 | Harper (31<sup>st</sup> Edn): 286 | DM Vasudevan (9<sup>th</sup> Edn): 306-310, 311

2. Write in detail about the protein biosynthesis in the body. Add a note on post translational modifications? [7 + 3 = 10]

Satyanarayana (5<sup>th</sup> Edn): 549 | Harper (31<sup>st</sup> Edn): 134, 406 | DM Vasudevan (9<sup>th</sup> Edn): 613, 619**WRITE SHORT NOTE ON****[5 × 4 = 20]**

3. Adrenal function test

Satyanarayana (5<sup>th</sup> Edn): 460 | Harper (31<sup>st</sup> Edn): 567

4. Biochemical actions of prostaglandins

Satyanarayana (5<sup>th</sup> Edn): 302, 618 | Harper (31<sup>st</sup> Edn): 224, 197

5. Mutations

Satyanarayana (5<sup>th</sup> Edn): 534 | Harper (31<sup>st</sup> Edn): 350-351 | DM Vasudevan (9<sup>th</sup> Edn): 630

6. Regulation of iron absorption

Satyanarayana (5<sup>th</sup> Edn): 136, 416 | Harper (31<sup>st</sup> Edn): 101, 523, 54 | DM Vasudevan (9<sup>th</sup> Edn): 537

7. Biochemical functions of zinc and iodine.

Satyanarayana (5<sup>th</sup> Edn): 420, 440, 421 | Harper (31<sup>st</sup> Edn): 562 | DM Vasudevan (9<sup>th</sup> Edn): 542**WRITE BRIEFLY ON****[5 × 2 = 10]**

8. Functions of thyroxine

Satyanarayana (5<sup>th</sup> Edn): 349, 420, 440, 503 | Harper (31<sup>st</sup> Edn): 498, 497 | DM Vasudevan (9<sup>th</sup> Edn): 681

9. What are tumor markers? Give two examples.

Satyanarayana (5<sup>th</sup> Edn): 662 | Harper (31<sup>st</sup> Edn): 700-701 | DM Vasudevan (9<sup>th</sup> Edn): 715

10. Mention the normal serum and urinary levels of total protein and albumin.

Satyanarayana (5<sup>th</sup> Edn): 511 | Harper (31<sup>st</sup> Edn): 223

11. Nucleotides

Satyanarayana (5<sup>th</sup> Edn): 72, 272, 389 | Harper (31<sup>st</sup> Edn): 322 | DM Vasudevan (9<sup>th</sup> Edn): 576, 577

12. Respiratory alkalosis.

Satyanarayana (5<sup>th</sup> Edn): 484 | Harper (31<sup>st</sup> Edn): 275 | DM Vasudevan (9<sup>th</sup> Edn): 449

# M.B.B.S DEGREE EXAMINATION

December - 2015

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - I

Time: 2½ Hours

Note : Answer all questions : Illustrate your answer with suitable diagrams.

1. Describe the sources, dietary requirements, biochemical functions and deficiency manifestations of vitamin A. [4 + 4 + 2 = 10]  
*Satyanarayana (5<sup>th</sup> Edn): 122 | Harper (31<sup>st</sup> Edn): 528, 529 | DM Vasudevan (9<sup>th</sup> Edn): 488-492*
2. Describe the pathway of ketone bodies synthesis and utilization. Add a note on ketoacidosis. [6 + 4 = 10]  
*Satyanarayana (5<sup>th</sup> Edn): 297, 481, 654 | Harper (31<sup>st</sup> Edn): 207, 210 | DM Vasudevan (9<sup>th</sup> Edn): 213, 215*

### WRITE SHORT NOTE ON

[5 × 4 = 20]

3. Lactose intolerance.  
*Satyanarayana (5<sup>th</sup> Edn): 171 | Harper (31<sup>st</sup> Edn): 519-520 | DM Vasudevan (9<sup>th</sup> Edn): 128*
4. Isoenzymes.  
*Satyanarayana (5<sup>th</sup> Edn): 112, 114 | Harper (31<sup>st</sup> Edn): 66 | DM Vasudevan (9<sup>th</sup> Edn): 72*
5. Significance of Hexose Monophosphate Pathway (HMP shunt).  
*Satyanarayana (5<sup>th</sup> Edn): 269, 383 | Harper (31<sup>st</sup> Edn): 182-190 | DM Vasudevan (9<sup>th</sup> Edn): 147*
6. Jaundice.  
*Satyanarayana (5<sup>th</sup> Edn): 215, 456 | Harper (31<sup>st</sup> Edn): 305 | DM Vasudevan (9<sup>th</sup> Edn): 370-397*
7. Inhibitors of respiratory chain.  
*Satyanarayana (5<sup>th</sup> Edn): 230 | Harper (31<sup>st</sup> Edn): 118 | DM Vasudevan (9<sup>th</sup> Edn): 352-354*

### WRITE BRIEFLY ON

[5 × 2 = 10]

8. Functions of essential fatty acids.  
*Satyanarayana (5<sup>th</sup> Edn): 32, 508 | Harper (31<sup>st</sup> Edn): 208 | DM Vasudevan (9<sup>th</sup> Edn): 255*
9. Polysaccharides  
*Satyanarayana (5<sup>th</sup> Edn): 10, 20 | Harper (31<sup>st</sup> Edn): 142 | DM Vasudevan (9<sup>th</sup> Edn): 93*
10. Methemoglobin.  
*Satyanarayana (5<sup>th</sup> Edn): 20, 275 | Harper (31<sup>st</sup> Edn): 53-54, 649 | DM Vasudevan (9<sup>th</sup> Edn): 377*
11. Causes and clinical symptoms of kwashiorkor.  
*Satyanarayana (5<sup>th</sup> Edn): 183, 515 | Harper (31<sup>st</sup> Edn): 263-264, 524-525 | DM Vasudevan (9<sup>th</sup> Edn): 556*
12. Cori cycle.  
*Satyanarayana (5<sup>th</sup> Edn): 261 | Harper (31<sup>st</sup> Edn): 177, 178 | DM Vasudevan (9<sup>th</sup> Edn): 137*

# M.B.B.S DEGREE EXAMINATION

December - 2015

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - II

Max. Marks: 50

1. Describe in detail the biochemical functions of calcium in human body. Discuss the hormonal regulation of plasma calcium. [5 + 5 = 10]  
*Satyanarayana (5<sup>th</sup> Edn): 128, 208, 406 | Harper (31<sup>st</sup> Edn): 523, 507 | DM Vasudevan (9<sup>th</sup> Edn): 527-534*
2. Describe the synthesis of DNA in detail with suitable diagram. [10]  
*Satyanarayana (5<sup>th</sup> Edn): 525-583 | Harper (31<sup>st</sup> Edn): 341-342*

### WRITE SHORT NOTE ON

[5 × 4 = 20]

3. tRNA structure and function.  
*Satyanarayana (5<sup>th</sup> Edn): 83, 550 | Harper (31<sup>st</sup> Edn): 343, 374 | DM Vasudevan (9<sup>th</sup> Edn): 613, 614*
4. Structure and classes of Immunoglobulins.  
*Satyanarayana (5<sup>th</sup> Edn): 187, 575 | Harper (31<sup>st</sup> Edn): 639, 627, 643 | DM Vasudevan (9<sup>th</sup> Edn): 693, 694*
5. Outline the steps of synthesis of urea.  
*Satyanarayana (5<sup>th</sup> Edn): 338, 341 | Harper (31<sup>st</sup> Edn): 272 | DM Vasudevan (9<sup>th</sup> Edn): 272-275*
6. Blood buffers  
*Satyanarayana (5<sup>th</sup> Edn): 476*

7. Renal clearance tests.  
*Satyanarayana (5<sup>th</sup> Edn): 462 | Harper (31<sup>st</sup> Edn): 566 | DM Vasudevan (9<sup>th</sup> Edn): 414*

### WRITE BRIEFLY ON

[5 × 2 = 10]

8. Essential amino acids.  
*Satyanarayana (5<sup>th</sup> Edn): 509 | Harper (31<sup>st</sup> Edn): 264 | DM Vasudevan (9<sup>th</sup> Edn): 553*
9. What are oncogenes and give two examples.  
*Satyanarayana (5<sup>th</sup> Edn): 659 | Harper (31<sup>st</sup> Edn): 683-688 | DM Vasudevan (9<sup>th</sup> Edn): 711*
10. Write four causes for metabolic acidosis.  
*Satyanarayana (5<sup>th</sup> Edn): 483 | Harper (31<sup>st</sup> Edn): 275 | DM Vasudevan (9<sup>th</sup> Edn): 446*
11. Maple syrup urine disease.  
*Satyanarayana (5<sup>th</sup> Edn): 367 | Harper (31<sup>st</sup> Edn): 290, 293 | DM Vasudevan (9<sup>th</sup> Edn): 300*
12. Mechanisms of transport across biological membranes.  
*Satyanarayana (5<sup>th</sup> Edn): 624 | Harper (31<sup>st</sup> Edn): 467-472*

**M.B.B.S DEGREE EXAMINATION**

August - 2015

**First M.B.B.S Examination  
BIOCHEMISTRY****PAPER - I****M.B.B.S DEGREE EXAMINATION**

August - 2015

**First M.B.B.S Examination  
BIOCHEMISTRY****PAPER - II****Time: 2½ Hours****Note : Answer all questions : Illustrate your answer with suitable diagrams.****Max. Marks: 50**

1. Write the glycolysis pathway in red blood cells. Add a note on 2, 3 Bisphosphoglycerate formation and its importance. **[5 + 4 = 10]**

Sanyanarayana (5<sup>th</sup> Edn): 244, 251 | Harper (31<sup>st</sup> Edn): 157, 158 | DM Vasudevan (9<sup>th</sup> Edn): 139

2. Give an account of sources, chemistry, biochemical functions, deficiency diseases and daily requirement of vitamin D. **[2 + 2 + 2 + 2 = 10]**

Sanyanarayana (5<sup>th</sup> Edn): 127 | Harper (31<sup>st</sup> Edn): 528 | DM Vasudevan (9<sup>th</sup> Edn): 492-496**WRITE SHORT NOTE ON****[5 × 4 = 20]**

3. Structure and classification of lipoproteins.

Sanyanarayana (5<sup>th</sup> Edn): 319 | Harper (31<sup>st</sup> Edn): 237, 238 | DM Vasudevan (9<sup>th</sup> Edn): 224

4. Factors affecting enzyme activity.

Sanyanarayana (5<sup>th</sup> Edn): 91-93 | Harper (31<sup>st</sup> Edn): 75 | DM Vasudevan (9<sup>th</sup> Edn): 62-70

5. Fatty liver.

Sanyanarayana (5<sup>th</sup> Edn): 324 | Harper (31<sup>st</sup> Edn): 260, 261 | DM Vasudevan (9<sup>th</sup> Edn): 211, 212

6. Functions and deficiency manifestations of Thiamine.

Sanyanarayana (5<sup>th</sup> Edn): 138 | Harper (31<sup>st</sup> Edn): 528, 533 | DM Vasudevan (9<sup>th</sup> Edn): 502, 503

7. Absorption of Monosaccharides.

Sanyanarayana (5<sup>th</sup> Edn): 10, 15, 18 | Harper (31<sup>st</sup> Edn): 141-142 | DM Vasudevan (9<sup>th</sup> Edn): 128, 129, 130**WRITE BRIEFLY ON****[5 × 2 = 10]**

8. List the primary and secondary bile acids.

Sanyanarayana (5<sup>th</sup> Edn): 174, 315 | Harper (31<sup>st</sup> Edn): 255-256 | DM Vasudevan (9<sup>th</sup> Edn): 231

9. Caloric value.

Sanyanarayana (5<sup>th</sup> Edn): 501 | DM Vasudevan (9<sup>th</sup> Edn): 549

10. Write any four heteropolysaccharides.

Sanyanarayana (5<sup>th</sup> Edn): 10, 21, 23 | Harper (31<sup>st</sup> Edn): 145-147 | DM Vasudevan (9<sup>th</sup> Edn): 93

11. Steatorrhea.

Sanyanarayana (5<sup>th</sup> Edn): 177 | Harper (31<sup>st</sup> Edn): 496 | DM Vasudevan (9<sup>th</sup> Edn): 765

12. Define xenobiotics and give two examples.

Sanyanarayana (5<sup>th</sup> Edn): 613 | Harper (31<sup>st</sup> Edn): 556 | DM Vasudevan (9<sup>th</sup> Edn): 478

1. Write in detail about urea cycle. Add a note on urea cycle disorders. **[6 + 4 = 10]**

Sanyanarayana (5<sup>th</sup> Edn): 338 | Harper (31<sup>st</sup> Edn): 272 | DM Vasudevan (9<sup>th</sup> Edn): 272-275

2. Write in detail about renal function tests. **[10]**

Sanyanarayana (5<sup>th</sup> Edn): 460 | Harper (31<sup>st</sup> Edn): 566 | DM Vasudevan (9<sup>th</sup> Edn): 410**WRITE SHORT NOTE ON****[5 × 4 = 20]**

3. Respiratory acidosis.

Sanyanarayana (5<sup>th</sup> Edn): 483 | DM Vasudevan (9<sup>th</sup> Edn): 449

4. Lac operon.

Sanyanarayana (5<sup>th</sup> Edn): 562 | Harper (31<sup>st</sup> Edn): 411-412, 414 | DM Vasudevan (9<sup>th</sup> Edn): 636

5. Primary and secondary structure of proteins.

Sanyanarayana (5<sup>th</sup> Edn): 55, 61 | Harper (31<sup>st</sup> Edn): 20-21, 35-37 | DM Vasudevan (9<sup>th</sup> Edn): 35, 37

6. Alkaptonuria.

Sanyanarayana (5<sup>th</sup> Edn): 353 | Harper (31<sup>st</sup> Edn): 286 | DM Vasudevan (9<sup>th</sup> Edn): 311

7. Functions of iodine and fluoride.

Sanyanarayana (5<sup>th</sup> Edn): 420, 440, 423 | Harper (31<sup>st</sup> Edn): 492, 159 | DM Vasudevan (9<sup>th</sup> Edn): 542, 543**WRITE BRIEFLY ON****[5 × 2 = 10]**

8. Post transcriptional modifications.

Sanyanarayana (5<sup>th</sup> Edn): 544 | Harper (31<sup>st</sup> Edn): 15, 17 | DM Vasudevan (9<sup>th</sup> Edn): 611

9. Isoelectric p<sup>H</sup>.

Sanyanarayana (5<sup>th</sup> Edn): 52, 64, 69 | Harper (31<sup>st</sup> Edn): 19 | DM Vasudevan (9<sup>th</sup> Edn): 42

10. Active methionine.

Sanyanarayana (5<sup>th</sup> Edn): 345, 361 | Harper (31<sup>st</sup> Edn): 298 | DM Vasudevan (9<sup>th</sup> Edn): 288

11. Most commonly used tumor markers.

Sanyanarayana (5<sup>th</sup> Edn): 662 | Harper (31<sup>st</sup> Edn): 700-701 | DM Vasudevan (9<sup>th</sup> Edn): 715

12. Normal serum levels of sodium, potassium, chloride and bicarbonate.

DM Vasudevan (9<sup>th</sup> Edn): 461, 462, 413

# M.B.B.S DEGREE EXAMINATION

November - 2014

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - I

Time: 2½ Hours

N T R U H S

# M.B.B.S DEGREE EXAMINATION

November - 2014

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - II

Note : Answer all questions : Illustrate your answer with suitable diagrams.

Max. Marks: 50

1. Describe the process of  $\beta$  (beta) – oxidation of fatty acids. [7 + 3 = 10]  
Add a note on the energetics of the pathway.

*Sanyanarayana (5<sup>th</sup> Edn): 287, 381 | Harper (31<sup>st</sup> Edn): 131 | DM Vasudevan (9<sup>th</sup> Edn): 201, 202, 203*

2. A 6 year boy was taken to the hospital by his mother with complaints of decreased vision in the night. The doctor suspected a possible vitamin A deficiency. Describe in detail the sources, RDA, function and deficiency manifestation of the deficient nutrient. What other clinical features, the doctor has to look for in this case and what advice should be given? [1+1+3+3+2=10]  
*Sanyanarayana (5<sup>th</sup> Edn): 119, 122 | Harper (31<sup>st</sup> Edn): 529-530 | DM Vasudevan (9<sup>th</sup> Edn): 488-492*

### WRITE SHORT NOTE ON

3. Describe briefly the hormonal regulation of blood glucose levels. [5 x 4 = 20]  
*Sanyanarayana (5<sup>th</sup> Edn): 241, 640 | Harper (31<sup>st</sup> Edn): 178-180 | DM Vasudevan (9<sup>th</sup> Edn): 135*

4. Draw a plot of competitive enzyme inhibition. Give 2 examples of competitive inhibitors.

5. Briefly explain the following:  
*Sanyanarayana (5<sup>th</sup> Edn): 95 | Harper (31<sup>st</sup> Edn): 76, 77 | DM Vasudevan (9<sup>th</sup> Edn): 65, 66*

(i) Specific Dynamic Action (ii) Glycemic index

6. Describe the clinical significance of the following enzymes:  
*Sanyanarayana (5<sup>th</sup> Edn): 503, 506 | Harper (31<sup>st</sup> Edn): 145, 520 | DM Vasudevan (9<sup>th</sup> Edn): 551, 560*

(i) LDH (ii) Alkaline phosphatase

- (iii) Creatine kinase (iv) Amylase  
*Sanyanarayana (5<sup>th</sup> Edn): 147, 456, 460, 112, 114, 110, 112 | Harper (31<sup>st</sup> Edn): 237, 605, 620, 520, 272, 273 | DM Vasudevan (9<sup>th</sup> Edn): 73, 74, 239, 76, 74*

7. Give an outline of the electron transport chain including ATP generating sites.  
*Sanyanarayana (5<sup>th</sup> Edn): 227 | Harper (31<sup>st</sup> Edn): 118, 120 | DM Vasudevan (9<sup>th</sup> Edn): 346-354*

### WRITE BRIEFLY ON

8. Absolute specificity enzymes. [5 x 2 = 10]

9. Altered CNS behavior in patients with advanced liver disease.  
*Sanyanarayana (5<sup>th</sup> Edn): 98 | Harper (31<sup>st</sup> Edn): 58 | DM Vasudevan (9<sup>th</sup> Edn): 70*

10. Ethanol is administered in patients with methanol poisoning.

11. Mention functions of Hemoglobinopathies.  
*Sanyanarayana (5<sup>th</sup> Edn): 614 | Harper (31<sup>st</sup> Edn): 557 | DM Vasudevan (9<sup>th</sup> Edn): 567, 568*

12. Lactose intolerance.  
*Sanyanarayana (5<sup>th</sup> Edn): 57, 694 | Harper (31<sup>st</sup> Edn): 651, 53-54 | DM Vasudevan (9<sup>th</sup> Edn): 378*

13. Discuss the metabolism of glycine under the following headings.  
*Sanyanarayana (5<sup>th</sup> Edn): 47, 208, 343, 616 | Harper (31<sup>st</sup> Edn): 265, 268 | DM Vasudevan (9<sup>th</sup> Edn): 282, 283*

14. Give an account of post transcription modifications.  
*Sanyanarayana (5<sup>th</sup> Edn): 344 | Harper (31<sup>st</sup> Edn): 15, 17 | DM Vasudevan (9<sup>th</sup> Edn): 611*

15. Describe PCR with diagram and list out two uses of it.  
*Sanyanarayana (5<sup>th</sup> Edn): 579 | Harper (31<sup>st</sup> Edn): 439 | DM Vasudevan (9<sup>th</sup> Edn): 653*

16. Compare and contrast (1 similarity and 2 differences) DNA polymerase and RNA polymerase.  
*Sanyanarayana (5<sup>th</sup> Edn): 525-527, 540, 543 | Harper (31<sup>st</sup> Edn): 341, 342 | DM Vasudevan (9<sup>th</sup> Edn): 598, 609*

17. Compare and contrast (1 similarity and 2 differences) prehepatic and post hepatic jaundice.  
*Sanyanarayana (5<sup>th</sup> Edn): 215, 456 | Harper (31<sup>st</sup> Edn): 566 | DM Vasudevan (9<sup>th</sup> Edn): 397*

18. Azaserine is used as an anticancer agent.  
*Sanyanarayana (5<sup>th</sup> Edn): 328*

19. Wilson's disease.  
*Sanyanarayana (5<sup>th</sup> Edn): 420 | Harper (31<sup>st</sup> Edn): 298, 627 | DM Vasudevan (9<sup>th</sup> Edn): 438, 439*

20. Oxalates in diet inhibit absorption of iron and calcium.  
*Sanyanarayana (5<sup>th</sup> Edn): 406, 414 | DM Vasudevan (9<sup>th</sup> Edn): 528, 537 | DM Vasudevan (9<sup>th</sup> Edn): 440*

21. Serum alpha fetoprotein and carcinoembryonic antigen.  
*Sanyanarayana (5<sup>th</sup> Edn): 663 | DM Vasudevan (9<sup>th</sup> Edn): 725*

1. A 15 year old boy complained of swelling and pain in the distal phalangeal joints. Blood investigation shown the following results.  
Blood urea : 15 mg%; Serum uric acid : 16 mg%  
On diagnosing the pathology, the physician decided to treat patient with allopurinol.

- (i) What is your probable diagnosis in the above patient?  
(ii) Which 2 other blood investigations would you suggest?  
(iii) Comment on the serum uric acid level and explain the cause of pain in joints.  
(iv) What is the biochemical explanation for the treatment given in the above patient? [1 + 1 + 4 + 4 = 10]  
*Sanyanarayana (5<sup>th</sup> Edn): 394, 396*

2. Discuss the metabolism of glycine under the following headings.  
Headings :  
(a) Synthesis (b) Catabolism (c) Specialized compounds synthesized  
*Sanyanarayana (5<sup>th</sup> Edn): 47, 208, 343, 616 | Harper (31<sup>st</sup> Edn): 265, 268 | DM Vasudevan (9<sup>th</sup> Edn): 282, 283*

3. Explain the absorption, transport and storage of dietary iron in the body. [5 x 4 = 20]  
*Sanyanarayana (5<sup>th</sup> Edn): 156, 416 | Harper (31<sup>st</sup> Edn): 523-524, 632-633 | DM Vasudevan (9<sup>th</sup> Edn): 536, 537, 538*

4. Give an account of post transcription modifications.  
*Sanyanarayana (5<sup>th</sup> Edn): 344 | Harper (31<sup>st</sup> Edn): 15, 17 | DM Vasudevan (9<sup>th</sup> Edn): 611*

5. Describe PCR with diagram and list out two uses of it.  
*Sanyanarayana (5<sup>th</sup> Edn): 579 | Harper (31<sup>st</sup> Edn): 439 | DM Vasudevan (9<sup>th</sup> Edn): 653*

6. Compare and contrast (1 similarity and 2 differences) DNA polymerase and RNA polymerase.  
*Sanyanarayana (5<sup>th</sup> Edn): 525-527, 540, 543 | Harper (31<sup>st</sup> Edn): 341, 342 | DM Vasudevan (9<sup>th</sup> Edn): 598, 609*

7. Compare and contrast (1 similarity and 2 differences) prehepatic and post hepatic jaundice.  
*Sanyanarayana (5<sup>th</sup> Edn): 215, 456 | Harper (31<sup>st</sup> Edn): 566 | DM Vasudevan (9<sup>th</sup> Edn): 397*

8. Azaserine is used as an anticancer agent.  
*Sanyanarayana (5<sup>th</sup> Edn): 328*

9. Wilson's disease.  
*Sanyanarayana (5<sup>th</sup> Edn): 420 | Harper (31<sup>st</sup> Edn): 298, 627 | DM Vasudevan (9<sup>th</sup> Edn): 438, 439*

10. Oxalates in diet inhibit absorption of iron and calcium.  
*Sanyanarayana (5<sup>th</sup> Edn): 406, 414 | DM Vasudevan (9<sup>th</sup> Edn): 528, 537 | DM Vasudevan (9<sup>th</sup> Edn): 440*

11. Serum alpha fetoprotein and carcinoembryonic antigen.  
*Sanyanarayana (5<sup>th</sup> Edn): 663 | DM Vasudevan (9<sup>th</sup> Edn): 725*

### WRITE BRIEFLY ON

8. Azaserine is used as an anticancer agent.  
*Sanyanarayana (5<sup>th</sup> Edn): 328*

9. Wilson's disease.  
*Sanyanarayana (5<sup>th</sup> Edn): 420 | Harper (31<sup>st</sup> Edn): 298, 627 | DM Vasudevan (9<sup>th</sup> Edn): 438, 439*

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[5 x 2 = 10]

**M.B.B.S DEGREE EXAMINATION***July/August - 2014***First M.B.B.S Examination****BIOCHEMISTRY****PAPER - I****M.B.B.S DEGREE EXAMINATION***July/August - 2014***First M.B.B.S Examination****BIOCHEMISTRY****PAPER - II****Time: 2½ Hours****Note : Answer all questions : Illustrate your answer with suitable diagrams.****Max. Marks: 50**

1. What are the different types of lipoproteins? Discuss their role in cholesterol transport. **[5 + 5 = 10]**

*Satyanarayana (5<sup>th</sup> Edn): 319, 320 | Harper (31<sup>st</sup> Edn): 258, 133, 628 | DM Vasudevan (9<sup>th</sup> Edn): 224*

2. Describe the tricarboxylic acid cycle and explain its significance. Add a note on the energetics. **[3 + 3 + 4 = 10]**

*Satyanarayana (5<sup>th</sup> Edn): 253, 383 | Harper (31<sup>st</sup> Edn): 150 | DM Vasudevan (9<sup>th</sup> Edn): 334***WRITE SHORT NOTE ON****[5 × 4 = 20]**

3. Porphyrias.

*Satyanarayana (5<sup>th</sup> Edn): 210 | Harper (31<sup>st</sup> Edn): 305 | DM Vasudevan (9<sup>th</sup> Edn): 363*

4. Competitive inhibition with examples.

*Satyanarayana (5<sup>th</sup> Edn): 94 | Harper (31<sup>st</sup> Edn): 76-78, 77 | DM Vasudevan (9<sup>th</sup> Edn): 65, 66*

5. Calorific value.

*Satyanarayana (5<sup>th</sup> Edn): 501 | DM Vasudevan (9<sup>th</sup> Edn): 549*

6. Detoxification by conjugation.

*Satyanarayana (5<sup>th</sup> Edn): 344 | Harper (31<sup>st</sup> Edn): 558 | DM Vasudevan (9<sup>th</sup> Edn): 480*

7. Function of Vitamin C.

*Satyanarayana (5<sup>th</sup> Edn): 135, 276 | Harper (31<sup>st</sup> Edn): 539 | DM Vasudevan (9<sup>th</sup> Edn): 519, 520, 521***WRITE BRIEFLY ON****[5 × 2 = 10]**

8. Essential fatty acids.

*Satyanarayana (5<sup>th</sup> Edn): 32, 508 | Harper (31<sup>st</sup> Edn): 196, 216 | DM Vasudevan (9<sup>th</sup> Edn): 255*

9. Sickle cell anemia.

*Satyanarayana (5<sup>th</sup> Edn): 201, 608 | Harper (31<sup>st</sup> Edn): 650 | DM Vasudevan (9<sup>th</sup> Edn): 379*

10. Dietary fibre and its role.

*Satyanarayana (5<sup>th</sup> Edn): 318, 507 | DM Vasudevan (9<sup>th</sup> Edn): 552*

11. Allosteric enzymes.

*Satyanarayana (5<sup>th</sup> Edn): 103 | Harper (31<sup>st</sup> Edn): 85 | DM Vasudevan (9<sup>th</sup> Edn): 68*

12. Define Epimers. Name two Epimers.

*Satyanarayana (5<sup>th</sup> Edn): 154 | Harper (31<sup>st</sup> Edn): 143*

1. What are the types of DNA? Describe the structure of B-DNA. How is DNA organized in the nucleolus? **[3 + 3 + 4 = 10]**

*Satyanarayana (5<sup>th</sup> Edn): 579 | Harper (31<sup>st</sup> Edn): 339*

2. Describe the various biochemical liver function tests. Explain the biochemical findings in different types of jaundice. **[5 + 5 = 10]**

*Satyanarayana (5<sup>th</sup> Edn): 454 | Harper (31<sup>st</sup> Edn): 566, 315 | DM Vasudevan (9<sup>th</sup> Edn): 397***WRITE SHORT NOTE ON****[5 × 4 = 20]**

3. Role of lungs in maintenance of body pH.

*Satyanarayana (5<sup>th</sup> Edn): 476 | DM Vasudevan (9<sup>th</sup> Edn): 441*

4. Creatinine clearance.

*Satyanarayana (5<sup>th</sup> Edn): 462 | Harper (31<sup>st</sup> Edn): 566 | DM Vasudevan (9<sup>th</sup> Edn): 415*

5. Polymerase chain reaction.

*Satyanarayana (5<sup>th</sup> Edn): 579 | Harper (31<sup>st</sup> Edn): 439-440 | DM Vasudevan (9<sup>th</sup> Edn): 658*

6. Role of parathormone in calcium homeostasis.

*Satyanarayana (5<sup>th</sup> Edn): 268, 406 | Harper (31<sup>st</sup> Edn): 532 | DM Vasudevan (9<sup>th</sup> Edn): 530, 531*

7. Tumor markers.

*Satyanarayana (5<sup>th</sup> Edn): 662 | Harper (31<sup>st</sup> Edn): 700-701 | DM Vasudevan (9<sup>th</sup> Edn): 715***WRITE BRIEFLY ON****[5 × 2 = 10]**

8. Anion gap.

*Satyanarayana (5<sup>th</sup> Edn): 482 | DM Vasudevan (9<sup>th</sup> Edn): 176, 446*

9. Mention four biochemical functions of zinc.

*Satyanarayana (5<sup>th</sup> Edn): 421 | Harper (31<sup>st</sup> Edn): 98 | DM Vasudevan (9<sup>th</sup> Edn): 542*

10. Wobble hypothesis.

*Satyanarayana (5<sup>th</sup> Edn): 548 | Harper (31<sup>st</sup> Edn): 396 | DM Vasudevan (9<sup>th</sup> Edn): 615*

11. Oncogenes.

*Satyanarayana (5<sup>th</sup> Edn): 659 | Harper (31<sup>st</sup> Edn): 685-688 | DM Vasudevan (9<sup>th</sup> Edn): 711*

12. Maple syrup urine.

*Satyanarayana (5<sup>th</sup> Edn): 367 | Harper (31<sup>st</sup> Edn): 286 | DM Vasudevan (9<sup>th</sup> Edn): 300*

# M.B.B.S DEGREE EXAMINATION

November - 2013

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - I

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# M.B.B.S DEGREE EXAMINATION

November - 2013

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - II

Time: 2½ Hours

Note : Answer all questions : Illustrate your answer with suitable diagrams.

Max. Marks: 50

1. Explain the components of electron transport chain and the flow of electron through them. Add a note on inhibitors and uncouplers.

[3 + 3 + 2 + 2 = 10]

2. Define 'glycogenesis' and 'glycogenolysis'. Describe glycogenesis in detail. How is it regulated?

[2 + 5 + 3 = 10]

Satyanarayana (5<sup>th</sup> Edn): 224-230 | Harper (31<sup>st</sup> Edn): 118, 120 | DM Vasudevan (9<sup>th</sup> Edn): 346, 143, 352, 354

### WRITE SHORT NOTE ON

[5 × 4 = 20]

3. Sickle cell haemoglobin.

Satyanarayana (5<sup>th</sup> Edn): 201 | Harper (31<sup>st</sup> Edn): 337 | DM Vasudevan (9<sup>th</sup> Edn): 379

4. Therapeutic enzymes.

Satyanarayana (5<sup>th</sup> Edn): 105 | Harper (31<sup>st</sup> Edn): 68 | DM Vasudevan (9<sup>th</sup> Edn): 76, 77

5. Functions of cholesterol.

Satyanarayana (5<sup>th</sup> Edn): 310 | Harper (31<sup>st</sup> Edn): 202 | DM Vasudevan (9<sup>th</sup> Edn): 221

6. Wald's visual cycle.

Satyanarayana (5<sup>th</sup> Edn): 124 | DM Vasudevan (9<sup>th</sup> Edn): 489

7. Biochemical functions of vitamin B<sub>12</sub>.

Satyanarayana (5<sup>th</sup> Edn): 156, 293, 422 | Harper (31<sup>st</sup> Edn): 536, 528 | DM Vasudevan (9<sup>th</sup> Edn): 516

### WRITE BRIEFLY ON

[5 × 2 = 10]

8. Metalloenzymes.

Satyanarayana (5<sup>th</sup> Edn): 93 | Harper (31<sup>st</sup> Edn): 58 | DM Vasudevan (9<sup>th</sup> Edn): 56

9. Acute intermittent porphyria.

Satyanarayana (5<sup>th</sup> Edn): 210 | Harper (31<sup>st</sup> Edn): 310, 311 | DM Vasudevan (9<sup>th</sup> Edn): 364

10. Pyruvate dehydrogenase complex.

Satyanarayana (5<sup>th</sup> Edn): 138, 252 | Harper (31<sup>st</sup> Edn): 161 | DM Vasudevan (9<sup>th</sup> Edn): 138

11. Liposomes.

Satyanarayana (5<sup>th</sup> Edn): 42 | Harper (31<sup>st</sup> Edn): 204 | DM Vasudevan (9<sup>th</sup> Edn): 105

12. Specific dynamic action.

Satyanarayana (5<sup>th</sup> Edn): 503 | Harper (31<sup>st</sup> Edn): 156 | DM Vasudevan (9<sup>th</sup> Edn): 551

1. Explain how recombinant DNA is formed. What are the application of recombinant DNA technology in medicine?

[5 + 5 = 10]

Satyanarayana (5<sup>th</sup> Edn): 532, 578 | Harper (31<sup>st</sup> Edn): 361-369 | DM Vasudevan (9<sup>th</sup> Edn): 643-647

2. Describe the formation of uric acid. What is the normal serum uric acid level? Explain the disease associated with its accumulation. Write the ways for lowering serum uric acid level.

[4 + 1 + 3 + 2 = 10]

Satyanarayana (5<sup>th</sup> Edn): 395 | Harper (31<sup>st</sup> Edn): 321 | DM Vasudevan (9<sup>th</sup> Edn): 582, 583

### WRITE SHORT NOTE ON

[5 × 4 = 20]

3. Urea cycle.

Satyanarayana (5<sup>th</sup> Edn): 338 | Harper (31<sup>st</sup> Edn): 272 | DM Vasudevan (9<sup>th</sup> Edn): 272-275

4. Nucleosomes.

Satyanarayana (5<sup>th</sup> Edn): 81 | Harper (31<sup>st</sup> Edn): 351 | DM Vasudevan (9<sup>th</sup> Edn): 594, 595

5. Structure of collagen.

Satyanarayana (5<sup>th</sup> Edn): 62, 67, 136 | Harper (31<sup>st</sup> Edn): 406 | DM Vasudevan (9<sup>th</sup> Edn): 721

6. Fluid mosaic model.

Satyanarayana (5<sup>th</sup> Edn): 623 | Harper (31<sup>st</sup> Edn): 465 | DM Vasudevan (9<sup>th</sup> Edn): 15

7. Functions of albumin in the body.

Satyanarayana (5<sup>th</sup> Edn): 623, 634, 671-675, 668, 669 | Harper (31<sup>st</sup> Edn): 599 | DM Vasudevan (9<sup>th</sup> Edn): 427

### WRITE BRIEFLY ON

[5 × 2 = 10]

8. Functions of Thyroid stimulating hormone.

Satyanarayana (5<sup>th</sup> Edn): 436, 441 | Harper (31<sup>st</sup> Edn): 567 | DM Vasudevan (9<sup>th</sup> Edn): 674, 675

9. Reverse transcriptase.

Satyanarayana (5<sup>th</sup> Edn): 546, 667 | Harper (31<sup>st</sup> Edn): 344, 360

10. Biochemical functions of zinc in the body.

Satyanarayana (5<sup>th</sup> Edn): 421 | Harper (31<sup>st</sup> Edn): 98, 99 | DM Vasudevan (9<sup>th</sup> Edn): 542

11. Oncogenes.

Satyanarayana (5<sup>th</sup> Edn): 659 | Harper (31<sup>st</sup> Edn): 688 | DM Vasudevan (9<sup>th</sup> Edn): 711

12. Biochemical function of estrogens.

Satyanarayana (5<sup>th</sup> Edn): 316, 443, 447 | Harper (31<sup>st</sup> Edn): 470 | DM Vasudevan (9<sup>th</sup> Edn): 650

12. Specific dynamic action.

Satyanarayana (5<sup>th</sup> Edn): 503 | Harper (31<sup>st</sup> Edn): 156 | DM Vasudevan (9<sup>th</sup> Edn): 551

## M.B.B.S DEGREE EXAMINATION

July - 2013

First M.B.B.S Examination

BIOCHEMISTRY

PAPER - I

Time: 2½ Hours

1. Explain the reactions of citric acid cycle emphasizing energy yield and amphibolic use of intermediates. Add a note on the energetics. [6 + 4 = 10]

Satyanarayana (5<sup>th</sup> Edn): 253, 283 | Harper (31<sup>st</sup> Edn): 130 | DM Vasudevan (9<sup>th</sup> Edn): 334

2. Describe the ketogenesis pathway and explain its significance. What are the tests done to detect the presence of ketone bodies in urine? [2 + 2 + 1 + 2 + 2 = 10]

Satyanarayana (5<sup>th</sup> Edn): 295, 654 | Harper (31<sup>st</sup> Edn): 133 | DM Vasudevan (9<sup>th</sup> Edn): 215, 215

### WRITE SHORT NOTE ON

3. Both cellulose and starch are homopolysaccharides of glucose yet humans digest only starch. Explain why? [5 × 4 = 20]

Satyanarayana (5<sup>th</sup> Edn): 223, 507 | DM Vasudevan (9<sup>th</sup> Edn): 93, 94

4. Porphyrias.

Satyanarayana (5<sup>th</sup> Edn): 210 | Harper (31<sup>st</sup> Edn): 305 | DM Vasudevan (9<sup>th</sup> Edn): 363

5. Type I glycogen storage disease.

Satyanarayana (5<sup>th</sup> Edn): 269 | Harper (31<sup>st</sup> Edn): 166, 167 | DM Vasudevan (9<sup>th</sup> Edn): 147

6. Serum lipoproteins.

Satyanarayana (5<sup>th</sup> Edn): 319 | Harper (31<sup>st</sup> Edn): 238 | DM Vasudevan (9<sup>th</sup> Edn): 224

7. High energy compounds.

Satyanarayana (5<sup>th</sup> Edn): 220 | DM Vasudevan (9<sup>th</sup> Edn): 345

### WRITE BRIEFLY ON

8. Explain why Thiamine deficiency is one of the causes for lactic acidosis. [5 × 2 = 10]

Satyanarayana (5<sup>th</sup> Edn): 138 | Harper (31<sup>st</sup> Edn): 534 | DM Vasudevan (9<sup>th</sup> Edn): 503

9. Effect of temperature on enzyme activity.

Satyanarayana (5<sup>th</sup> Edn): 91 | Harper (31<sup>st</sup> Edn): 75 | DM Vasudevan (9<sup>th</sup> Edn): 65

10. Sources and deficiency manifestations of Vit.A.

Satyanarayana (5<sup>th</sup> Edn): 122 | Harper (31<sup>st</sup> Edn): 530, 528 | DM Vasudevan (9<sup>th</sup> Edn): 498

11. Proenzymes.

Satyanarayana (5<sup>th</sup> Edn): 105, 171 | Harper (31<sup>st</sup> Edn): 87, 88

12. Name different types of α-Thalassemias.

Satyanarayana (5<sup>th</sup> Edn): 205 | Harper (31<sup>st</sup> Edn): 54 | DM Vasudevan (9<sup>th</sup> Edn): 381

12. Biochemical function of estrogens.

Satyanarayana (5<sup>th</sup> Edn): 316, 443, 447 | Harper (31<sup>st</sup> Edn): 470 | DM Vasudevan (9<sup>th</sup> Edn): 680

## M.B.B.S DEGREE EXAMINATION

July - 2013

First M.B.B.S Examination

BIOCHEMISTRY

PAPER - II

Max. Marks: 50

Note : Answer all questions : Illustrate your answer with suitable diagrams.

1. With the help of a diagram, describe the process of transcription. Add a note on post transcriptional modifications. [6 + 4 = 10]

Satyanarayana (5<sup>th</sup> Edn): 544 | Harper (31<sup>st</sup> Edn): 341 | DM Vasudevan (9<sup>th</sup> Edn): 608-613

2. Enumerate kidney function tests and liver function tests. Write in detail about one kidney function test and one liver function test. [2 + 2 + 3 + 3 = 10]

Satyanarayana (5<sup>th</sup> Edn): 460 | Harper (31<sup>st</sup> Edn): 566 | DM Vasudevan (9<sup>th</sup> Edn): 397

### WRITE SHORT NOTE ON

3. Explain the absorption, transport and storage of dietary iron in the body. [5 × 4 = 20]

Satyanarayana (5<sup>th</sup> Edn): 136, 416 | Harper (31<sup>st</sup> Edn): 523-524, 632-634 | DM Vasudevan (9<sup>th</sup> Edn): 536, 537, 538

4. Requirement, sources and biochemical functions of selenium.

Satyanarayana (5<sup>th</sup> Edn): 132, 423 | Harper (31<sup>st</sup> Edn): 204, 244

5. Describe polymerase chain reaction (PCR) with diagram and list out two uses of it.

Satyanarayana (5<sup>th</sup> Edn): 579 | Harper (31<sup>st</sup> Edn): 439-440 | DM Vasudevan (9<sup>th</sup> Edn): 658

6. Compare and contrast (1 similarity and 2 difference) DNA polymerase and RNA polymerase.

Satyanarayana (5<sup>th</sup> Edn): 525, 527, 540, 543 | Harper (31<sup>st</sup> Edn): 364, 377 | DM Vasudevan (9<sup>th</sup> Edn): 598, 609

7. Androgens – Biosynthesis and physiological and – Biochemical functions.

Satyanarayana (5<sup>th</sup> Edn): 447 | Harper (31<sup>st</sup> Edn): 486 | DM Vasudevan (9<sup>th</sup> Edn): 680

8. Chemical Carcinogens. [5 × 2 = 10]

Satyanarayana (5<sup>th</sup> Edn): 658 | Harper (31<sup>st</sup> Edn): 556-557, 559, 683 | DM Vasudevan (9<sup>th</sup> Edn): 708, 709

9. Biochemical function of oxytocin.

Satyanarayana (5<sup>th</sup> Edn): 69, 439 | DM Vasudevan (9<sup>th</sup> Edn): 673

10. How oxalates in diet inhibit absorption of iron and calcium?

Satyanarayana (5<sup>th</sup> Edn): 136, 410, 128, 406 | DM Vasudevan (9<sup>th</sup> Edn): 528, 537

11. Define osmolality and Osmolality.

Satyanarayana (5<sup>th</sup> Edn): 472 | DM Vasudevan (9<sup>th</sup> Edn): 456

12. Serum alpha fetoprotein and carcinoembryonic antigen.

Satyanarayana (5<sup>th</sup> Edn): 663 | DM Vasudevan (9<sup>th</sup> Edn): 715

## M.B.B.S DEGREE EXAMINATION

November - 2012

First M.B.B.S Examination

### BIOCHEMISTRY

PAPER - I

Time: 2½ Hours

1. Describe the biochemical changes in carbohydrate and lipid metabolism in prolonged starvation.  
[3 + 3 + 2 + 2 = 10]  
*Satyannarayana (5th Edn): 355 | Harper (31st Edn): 242 | DM Vasudevan (9th Edn): 124*
2. Describe the formation, transport and metabolic rate of ammonia in the body. What is ammonia toxicity?  
[2 + 2 + 1 + 2 + 2 = 10]  
*Satyannarayana (5th Edn): 336 | Harper (31st Edn): 272 | DM Vasudevan (9th Edn): 272-275*

#### WRITE SHORT NOTE ON

3. Isoenzymes and their clinical importance.  
[5 × 4 = 20]

4. Folate trap (Methyl trap)  
*Satyannarayana (5th Edn): 112, 114 | Harper (31st Edn): 61-62 | DM Vasudevan (9th Edn): 72*

5. Define basal metabolic rate (BMR) and list the factors affecting BMR.  
*Satyannarayana (5th Edn): 160 | Harper (31st Edn): 538 | DM Vasudevan (9th Edn): 517*

6. Advantages and disadvantages of intake of polyunsaturated fatty acids.  
*Satyannarayana (5th Edn): 502 | Harper (31st Edn): 524 | DM Vasudevan (9th Edn): 558*

7. Structure and function of glutathione.  
*Satyannarayana (5th Edn): 32, 132, 317, 508 | Harper (31st Edn): 220 | DM Vasudevan (9th Edn): 254, 255*

8. Dietary fibre.  
*Satyannarayana (5th Edn): 173, 344 | Harper (31st Edn): 713, 721*

9. Two main causes of Fatty liver.  
*Satyannarayana (5th Edn): 169, 508 | DM Vasudevan (9th Edn): 553*

#### WRITE BRIEFLY ON

[5 × 2 = 10]

10. Main sources of sucrose and lactose.  
*Satyannarayana (5th Edn): 324 | Harper (31st Edn): 244 | DM Vasudevan (9th Edn): 211*
11. Mention four Biochemical functions of vitamin K.  
*Satyannarayana (5th Edn): 191, 277, 562 | Harper (31st Edn): 145, 146 | DM Vasudevan (9th Edn): 92*
12. Any four functions of phospholipids.  
*Satyannarayana (5th Edn): 133 | Harper (31st Edn): 532 | DM Vasudevan (9th Edn): 498*

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## M.B.B.S DEGREE EXAMINATION

November - 2012

First M.B.B.S Examination

### BIOCHEMISTRY

PAPER - II

Max. Marks: 50

1. Describe the process of DNA replication. Enumerate the DNA repair mechanisms.  
[8 + 2 = 10]  
*Satyannarayana (5th Edn): 525-528 | Harper (31st Edn): 341 | DM Vasudevan (9th Edn): 608-613*

2. Write the causes and biochemical findings in metabolic acidosis. Explain the compensatory mechanism.  
[2 + 3 + 5 = 10]  
*Satyannarayana (5th Edn): 483 | Harper (31st Edn): 275 | DM Vasudevan (9th Edn): 446, 447*

#### WRITE SHORT NOTE ON

3. Gout.  
[5 × 4 = 20]  
*Satyannarayana (5th Edn): 395, 398 | Harper (31st Edn): 190, 244 | DM Vasudevan (9th Edn): 583*

4. Gastric function tests.  
*Satyannarayana (5th Edn): 463 | DM Vasudevan (9th Edn): 404*

5. Structure of plasma membrane.  
*Satyannarayana (5th Edn): 623 | Harper (31st Edn): 459 | DM Vasudevan (9th Edn): 15*

6. Creatinine clearance test.  
*Satyannarayana (5th Edn): 462 | Harper (31st Edn): 455 | DM Vasudevan (9th Edn): 415*

7. Biochemical functions of copper in the body.  
*Satyannarayana (5th Edn): 93, 419 | Harper (31st Edn): 98-100 | DM Vasudevan (9th Edn): 541*

8. Mention the essential component (rare element) and precursor of T<sub>3</sub> & T<sub>4</sub>.  
[5 × 2 = 10]  
*Satyannarayana (5th Edn): 440 | Harper (31st Edn): 498 | DM Vasudevan (9th Edn): 681*

9. Functions of t-RNA.  
*Satyannarayana (5th Edn): 83, 550 | Harper (31st Edn): 343 | DM Vasudevan (9th Edn): 613, 614*

#### WRITE BRIEFLY ON

10. Name the four distinct phases of cell cycle.  
*Satyannarayana (5th Edn): 530 | Harper (31st Edn): 691 | DM Vasudevan (9th Edn): 632, 633*

11. Genetic code.  
*Satyannarayana (5th Edn): 547 | Harper (31st Edn): 338 | DM Vasudevan (9th Edn): 615*

12. Four factors affecting absorption of iron.  
*Satyannarayana (5th Edn): 136, 416 | Harper (31st Edn): 523 | DM Vasudevan (9th Edn): 537*

Time: 2½ Hours

Note : Answer all questions : Illustrate your answer with suitable diagrams.

1. Explain the reactions of Glycogenesis and Glycogenolysis in liver. How are these pathways regulated? [4 + 3 + 3 = 10]  
Sanyal (5th Edn): 263, 265 | Harper (31st Edn): 164 | DM Vasudevan (9th Edn): 143, 144
2. Give an account of the sources, chemistry, biochemical functions, deficiency diseases and daily requirement of vitamin A. [1 + 2 + 3 + 3 + 1 = 10]  
Sanyal (5th Edn): 122 | Harper (31st Edn): 529-530 | DM Vasudevan (9th Edn): 448-492

**WRITE SHORT NOTE ON**

3. Ketogenesis. [5 × 4 = 20]  
Sanyal (5th Edn): 295, 654 | Harper (31st Edn): 133 | DM Vasudevan (9th Edn): 213
4. Porphyrins.  
Sanyal (5th Edn): 210 | Harper (31st Edn): 305, 310 | DM Vasudevan (9th Edn): 363
5. Competitive inhibition.  
Sanyal (5th Edn): 94 | Harper (31st Edn): 76-78 | DM Vasudevan (9th Edn): 63, 66
6. Kwashiorkor and marasmus.  
Sanyal (5th Edn): 183, 515 | Harper (31st Edn): 263-264 | DM Vasudevan (9th Edn): 555, 556
7. Galactosemia.  
Sanyal (5th Edn): 278 | Harper (31st Edn): 191 | DM Vasudevan (9th Edn): 188

**WRITE BRIEFLY ON**

8. Essential fatty acids. [5 × 2 = 10]  
Sanyal (5th Edn): 32, 508 | Harper (31st Edn): 196, 216 | DM Vasudevan (9th Edn): 253
9. What are isomerases? Give two examples.  
Sanyal (5th Edn): 69 | Harper (31st Edn): 41 | DM Vasudevan (9th Edn): 55
10. Sickle cell haemoglobin.  
Sanyal (5th Edn): 201 | Harper (31st Edn): 397 | DM Vasudevan (9th Edn): 379
11. Enzyme defects in essential fructosuria and hereditary fructose intolerance.  
Sanyal (5th Edn): 280 | Harper (31st Edn): 190 | DM Vasudevan (9th Edn): 188, 187, 184, 328
12. Sources and functions of folic acid.  
Sanyal (5th Edn): 153 | Harper (31st Edn): 525 | DM Vasudevan (9th Edn): 512, 514

Max. Marks: 50

1. Explain the pathway for the degradation of purine nucleotides and enumerate the associated metabolic disorders [6 + 4 = 10]  
Sanyal (5th Edn): 75 | Harper (31st Edn): 327 | DM Vasudevan (9th Edn): 582, 583
2. Give an account of transamination, deamination and transmethylation. [4 + 3 + 3 = 10]  
Sanyal (5th Edn): 146, 333, 361, 334 | Harper (31st Edn): 131 | DM Vasudevan (9th Edn): 270, 271

**WRITE SHORT NOTE ON**

3. Regulation of plasma calcium. [5 × 4 = 20]  
Sanyal (5th Edn): 128, 268, 406 | Harper (31st Edn): 478 | DM Vasudevan (9th Edn): 529
4. Metabolic and respirator alkalosis.  
Sanyal (5th Edn): 483, 484 | Harper (31st Edn): 275 | DM Vasudevan (9th Edn): 443, 449
5. Plasma proteins and their functions.  
Sanyal (5th Edn): 182 | Harper (31st Edn): 133, 639 | DM Vasudevan (9th Edn): 425
6. Tumor markers.  
Sanyal (5th Edn): 662 | Harper (31st Edn): 700-701 | DM Vasudevan (9th Edn): 715

**Secondary structure of proteins.**

Sanyal (5th Edn): 55, 61 | Harper (31st Edn): 21 | DM Vasudevan (9th Edn): 57

**WRITE BRIEFLY ON**

8. Differences between amylose and amylopectin. [5 × 2 = 10]  
Sanyal (5th Edn): 21 | Harper (31st Edn): 145 | DM Vasudevan (9th Edn): 95, 94
9. What are conjugated proteins? Give two examples.  
Sanyal (5th Edn): 64 | Harper (31st Edn): 508 | DM Vasudevan (9th Edn): 44
10. Biochemical functions of sodium.  
Sanyal (5th Edn): 403 | Harper (31st Edn): 98, 99 | DM Vasudevan (9th Edn): 460, 461
11. Base pairing rule.  
Sanyal (5th Edn): 81 | Harper (31st Edn): 555, 554 | DM Vasudevan (9th Edn): 595
12. Alkaptonuria.  
Sanyal (5th Edn): 553 | Harper (31st Edn): 288 | DM Vasudevan (9th Edn): 511

# M.B.B.S DEGREE EXAMINATION

January - 2012

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - I

# M.B.B.S DEGREE EXAMINATION

January - 2012

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - II

Time: 2½ Hours

Note : Answer all questions : Illustrate your answer with suitable diagrams.

Max. Marks: 50

1. Explain the various factors affecting the enzyme activity. Add a note on the significance of the  $K_m$  value of the enzyme. [7 + 3 = 10]  
*Satyannarayana (5th Edn): 90-93 | Harper (31st Edn): 72-73 | DM Vasudevan (9th Edn): 62-70*
2. Discuss the mitochondrial electron transport chain (E.T.C) and the inhibitors at various sites of this chain. [10]  
*Satyannarayana (5th Edn): 224-235 | Harper (31st Edn): 118, 120 | DM Vasudevan (9th Edn): 343*

### WRITE SHORT NOTE ON

[5 × 4 = 20]

3. Detoxification by conjugation.  
*Satyannarayana (5th Edn): 344 | Harper (31st Edn): 558 | DM Vasudevan (9th Edn): 480*
4. Glycogen storage diseases.  
*Satyannarayana (5th Edn): 269 | Harper (31st Edn): 164 | DM Vasudevan (9th Edn): 147*
5. Fatty liver and lipotropic factors.  
*Satyannarayana (5th Edn): 324, 325 | Harper (31st Edn): 244 | DM Vasudevan (9th Edn): 211, 212*
6. Haemoglobinopathies with examples.  
*Satyannarayana (5th Edn): 201 | Harper (31st Edn): 53-54, 651 | DM Vasudevan (9th Edn): 378*
7. Functions and deficiency features of pyridoxine.  
*Satyannarayana (5th Edn): 146 | DM Vasudevan (9th Edn): 508*

### WRITE BRIEFLY ON

[5 × 2 = 10]

8. Define isoenzymes. Give two examples and their diagnostic use.  
*Satyannarayana (5th Edn): 112, 114 | Harper (31st Edn): 61-62 | DM Vasudevan (9th Edn): 72*
9. Functions of Bile Salts.  
*Satyannarayana (5th Edn): 174, 315 | Harper (31st Edn): 202, 249 | DM Vasudevan (9th Edn): 232, 231*
10. Energetics of TCA (tricarboxylic acid) cycle.  
*Satyannarayana (5th Edn): 253, 383 | Harper (31st Edn): 153 | DM Vasudevan (9th Edn): 337*
11. Define antioxidants. Which vitamins have antioxidant function?  
*Satyannarayana (5th Edn): 275, 632 | Harper (31st Edn): 1126 | DM Vasudevan (9th Edn): 476*
12. Respiratory Quotient.  
*Satyannarayana (5th Edn): 501 | DM Vasudevan (9th Edn): 549*

1. Describe the steps of transcription in Eukaryotes. Give two examples for post transcriptional modifications. [8 + 2 = 10]  
*Satyannarayana (5th Edn): 543 | Harper (31st Edn): 341 | DM Vasudevan (9th Edn): 608-613*
2. Discuss the metabolism of calcium under the following sections:  
(a) Sources  
(b) Recommended daily allowance  
(c) Functions (any 4)  
(d) Regulation of plasma level  
(e) Deficiency manifestations. [1 + 1 + 4 + 2 + 2 = 10]  
*Satyannarayana (5th Edn): 406 | Harper (31st Edn): 506, 523, 93 | DM Vasudevan (9th Edn): 527-534*
3. **WRITE SHORT NOTE ON**  
Plasma buffers and the role of buffers in the regulation of pH. [5 × 4 = 20]  
*Satyannarayana (5th Edn): 476, 678 | Harper (31st Edn): 12-13 | DM Vasudevan (9th Edn): 441, 442*
4. Mechanisms of action of hormones.  
*Satyannarayana (5th Edn): 430 | Harper (31st Edn): 484 | DM Vasudevan (9th Edn): 667, 668*
5. Active transport with 2 examples.  
*Satyannarayana (5th Edn): 627 | Harper (31st Edn): 467 | DM Vasudevan (9th Edn): 21*
6. Biological important products derived from glycine.  
*Satyannarayana (5th Edn): 342 | Harper (32nd Edn): 298, 300-301 | DM Vasudevan (9th Edn): 283*
7. Enumerate the liver function tests. Explain the detoxification of bilirubin by the liver.  
*Satyannarayana (5th Edn): 454 | Harper (31st Edn): 566 | DM Vasudevan (9th Edn): 397*
8. **WRITE BRIEFLY ON**  
Role of copper in iron metabolism. [5 × 2 = 10]  
*Satyannarayana (5th Edn): 136, 416 | Harper (31st Edn): 98-100 | DM Vasudevan (9th Edn): 537*
9. Orotic aciduria.  
*Satyannarayana (5th Edn): 402 | Harper (31st Edn): 336*
10. Bonds responsible for maintaining the higher levels of organization of protein structure.  
*Satyannarayana (5th Edn): 55, 61 | Harper (31st Edn): 33 | DM Vasudevan (9th Edn): 37*
11. Ribozymes.  
*Satyannarayana (5th Edn): 108, 553 | Harper (31st Edn): 66-67, 343 | DM Vasudevan (9th Edn): 612*
12. Oncofetal antigens.  
*Satyannarayana (5th Edn): 653 | DM Vasudevan (9th Edn): 774*

# M.B.B.S DEGREE EXAMINATION

July - 2011

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - I

M.T.R.U.H.S

# M.B.B.S DEGREE EXAMINATION

July - 2011

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - II

Time: 2½ Hours

Note : Answer all questions : Illustrate your answer with suitable diagrams.

Max. Marks: 50

1. Describe the CITRIC ACID cycle with energetics. Explain the Amphibolic role of this cycle? [6 + 2 + 2 = 10]

Sayanarayana (5<sup>th</sup> Edn): 253, 383 | Harper (31<sup>st</sup> Edn): 150-161 | DM Varadachari (9<sup>th</sup> Edn): 334

2. Write the sources, daily requirement, functions and deficiency manifestations of vitamin C? [2 + 1 + 4 + 3 = 10]

Sayanarayana (5<sup>th</sup> Edn): 135, 276 | Harper (31<sup>st</sup> Edn): 43, 263, 539 | DM Varadachari (9<sup>th</sup> Edn): 519-521

**WRITE SHORT NOTE ON** [5 x 4 = 20]

3. Chemiosmotic theory and oxidative phosphorylation.

Sayanarayana (5<sup>th</sup> Edn): 228, 226 | Harper (31<sup>st</sup> Edn): 121, 124 | DM Varadachari (9<sup>th</sup> Edn): 350-351

4. Porphyrins.

Sayanarayana (5<sup>th</sup> Edn): 210 | Harper (31<sup>st</sup> Edn): 305, 310 | DM Varadachari (9<sup>th</sup> Edn): 353

5. Digestion and absorption of lipids.

Sayanarayana (5<sup>th</sup> Edn): 173 | Harper (31<sup>st</sup> Edn): 521 | DM Varadachari (9<sup>th</sup> Edn): 198, 199, 200

6. Competitive inhibition and its clinical significance.

Sayanarayana (5<sup>th</sup> Edn): 94 | Harper (31<sup>st</sup> Edn): 76-78 | DM Varadachari (9<sup>th</sup> Edn): 65, 66

7. Enzyme markers in myocardial infarction.

Sayanarayana (5<sup>th</sup> Edn): 110 | Harper (31<sup>st</sup> Edn): 567 | DM Varadachari (9<sup>th</sup> Edn): 240

**WRITE BRIEFLY ON** [5 x 2 = 10]

8. Give two examples for phospholipids and write their functions.

Sayanarayana (5<sup>th</sup> Edn): 29, 35 | Harper (31<sup>st</sup> Edn): 196, 232 | DM Varadachari (9<sup>th</sup> Edn): 104, 405

9. Specific Dynamic Action.

Sayanarayana (5<sup>th</sup> Edn): 503 | Harper (31<sup>st</sup> Edn): 156 | 551

10. MUTAROTATION.

Sayanarayana (5<sup>th</sup> Edn): 15 | DM Varadachari (9<sup>th</sup> Edn): 83

11. What are Antivitamins? Give any two examples with vitamin inhibited by them.

Sayanarayana (5<sup>th</sup> Edn): 95, 163 | DM Varadachari (9<sup>th</sup> Edn): 569

12. Absolute specificity of Enzymes.

Sayanarayana (5<sup>th</sup> Edn): 98 | Harper (31<sup>st</sup> Edn): 58 | DM Varadachari (9<sup>th</sup> Edn): 70

1. What is the normal pH of blood? Discuss the respiratory and renal regulation of pH. [1 + 4 + 5 = 10]

Sayanarayana (5<sup>th</sup> Edn): 476 | Harper (31<sup>st</sup> Edn): 12-13 | DM Varadachari (9<sup>th</sup> Edn): 440, 441, 442

2. Explain the pathway of catabolism of Tyrosine. Write the important products synthesized from Tyrosine. Add a note on Albinism. [2 + 2 + 3 + 3 = 10]

Sayanarayana (5<sup>th</sup> Edn): 346, 354 | Harper (31<sup>st</sup> Edn): 206-208 | DM Varadachari (9<sup>th</sup> Edn): 506-508, 509

**WRITE SHORT NOTE ON** [5 x 4 = 20]

3. Replication.

Sayanarayana (5<sup>th</sup> Edn): 525-529 | Harper (31<sup>st</sup> Edn): 341 | DM Varadachari (9<sup>th</sup> Edn): 506-508

4. Fluid mosaic model of membrane structure.

Sayanarayana (5<sup>th</sup> Edn): 623 | Harper (31<sup>st</sup> Edn): 465 | DM Varadachari (9<sup>th</sup> Edn): 35

5. What is Recombinant DNA? What is the role of restriction endonuclease in recombinant DNA technique? [C16]

Sayanarayana (5<sup>th</sup> Edn): 571 | Harper (31<sup>st</sup> Edn): 362-369 | DM Varadachari (9<sup>th</sup> Edn): 643, 644, 647

6. Name the plasma proteins. List any four functions of them in the human body.

Sayanarayana (5<sup>th</sup> Edn): 187 | Harper (31<sup>st</sup> Edn): 133, 630 | DM Varadachari (9<sup>th</sup> Edn): 425

7. Creatinine Clearance test

Sayanarayana (5<sup>th</sup> Edn): 462 | Harper (31<sup>st</sup> Edn): 455 | DM Varadachari (9<sup>th</sup> Edn): 425

**WRITE BRIEFLY ON** [5 x 2 = 10]

8. Apoptosis.

Sayanarayana (5<sup>th</sup> Edn): 8, 639 | Harper (31<sup>st</sup> Edn): 710 | DM Varadachari (9<sup>th</sup> Edn): 774

9. Oxytocin.

Sayanarayana (5<sup>th</sup> Edn): 62, 439 | DM Varadachari (9<sup>th</sup> Edn): 673

10. FLUOROSIS.

Sayanarayana (5<sup>th</sup> Edn): 420 | DM Varadachari (9<sup>th</sup> Edn): 543

11. Vandenberg's test

Sayanarayana (5<sup>th</sup> Edn): 456 | DM Varadachari (9<sup>th</sup> Edn): 348

12. Anion gap.

Sayanarayana (5<sup>th</sup> Edn): 480 | DM Varadachari (9<sup>th</sup> Edn): 176, 446

# M.B.B.S DEGREE EXAMINATION

January - 2011

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - I

Time: 2½ Hours

Note : Answer all questions : Illustrate your answer with suitable diagrams.

1. Describe the sources, dietary requirements, biochemical functions and deficiency manifestations of vitamin A. [1 + 1 + 4 + 4 = 10]

Satyanarayana (5<sup>th</sup> Edn): 122 | Harper (31<sup>st</sup> Edn): 528, 530 | DM Vasudevan (9<sup>th</sup> Edn): 488-492

2. Mention the pathways by which glucose is metabolized in the body. Describe the steps of anaerobic glycolysis and its energetics. [2 + 6 + 2 = 10]

Satyanarayana (5<sup>th</sup> Edn): 244, 383 | Harper (31<sup>st</sup> Edn): 157 | DM Vasudevan (9<sup>th</sup> Edn): 128

### WRITE SHORT NOTE ON

[5 × 4 = 20]

3. Schematically represent the electron transport chain indicating the sites of ATP production and inhibitors.

Satyanarayana (5<sup>th</sup> Edn): 224-235 | Harper (31<sup>st</sup> Edn): 118, 120 | DM Vasudevan (9<sup>th</sup> Edn): 346-354

4. Formation and fact of bile pigments.

Satyanarayana (5<sup>th</sup> Edn): 214, 455 | Harper (31<sup>st</sup> Edn): 305

5. Factors affecting enzyme activity.

Satyanarayana (5<sup>th</sup> Edn): 90-93 | Harper (31<sup>st</sup> Edn): 75, 564-565 | DM Vasudevan (9<sup>th</sup> Edn): 62-70

6. Abnormal Haemoglobins.

Satyanarayana (5<sup>th</sup> Edn): 201 | Harper (31<sup>st</sup> Edn): 53, 54, 397 | DM Vasudevan (9<sup>th</sup> Edn): 378, 379

7. Lipoproteins.

Satyanarayana (5<sup>th</sup> Edn): 319 | Harper (31<sup>st</sup> Edn): 258, 628, 34 | DM Vasudevan (9<sup>th</sup> Edn): 224

### WRITE BRIEFLY ON

[5 × 2 = 10]

8. Biological value of proteins.

Satyanarayana (5<sup>th</sup> Edn): 511 | Harper (31<sup>st</sup> Edn): 519-526 | DM Vasudevan (9<sup>th</sup> Edn): 554

9. K<sub>m</sub> value and its significance.

Satyanarayana (5<sup>th</sup> Edn): 90, 94 | Harper (31<sup>st</sup> Edn): 75 | DM Vasudevan (9<sup>th</sup> Edn): 63, 64

10. Dietary fiber.

Satyanarayana (5<sup>th</sup> Edn): 169 | DM Vasudevan (9<sup>th</sup> Edn): 552

11. Detoxification by oxidation.

Satyanarayana (5<sup>th</sup> Edn): 613, 614 | Harper (31<sup>st</sup> Edn): 556 | DM Vasudevan (9<sup>th</sup> Edn): 479

12. Lactose Intolerance.

Satyanarayana (5<sup>th</sup> Edn): 171 | Harper (31<sup>st</sup> Edn): 519-520 | DM Vasudevan (9<sup>th</sup> Edn): 128

# M.B.B.S DEGREE EXAMINATION

January - 2011

First M.B.B.S Examination

## BIOCHEMISTRY

PAPER - II

Max. Marks: 50

Note : Answer all questions : Illustrate your answer with suitable diagrams.

1. Define a buffer. Explain the various mechanisms of regulation of acid-base balance. Add a note on acid-base disorders. [1 + 7 + 2 = 10]

Satyanarayana (5<sup>th</sup> Edn): 476 | Harper (31<sup>st</sup> Edn): 12-13 | DM Vasudevan (9<sup>th</sup> Edn): 439, 441, 442, 446

2. Describe the steps of eukaryotic replication of DNA. Add a note on Reverse transcription. [3 + 3 + 3 + 1 = 10]

Satyanarayana (5<sup>th</sup> Edn): 527 | Harper (31<sup>st</sup> Edn): 394, | DM Vasudevan (9<sup>th</sup> Edn): 600, 612

### WRITE SHORT NOTE ON

[5 × 4 = 20]

3. Absorption and transportation of Iron.

Satyanarayana (5<sup>th</sup> Edn): 136, 416 | Harper (31<sup>st</sup> Edn): 523-524, 632-633 | DM Vasudevan (9<sup>th</sup> Edn): 536, 537, 538

4. Purine salvage pathways.

Satyanarayana (5<sup>th</sup> Edn): 393 | Harper (31<sup>st</sup> Edn): 328 | DM Vasudevan (9<sup>th</sup> Edn): 582

5. Outline the steps of biosynthesis of Urea.

Satyanarayana (5<sup>th</sup> Edn): 338, 341 | Harper (31<sup>st</sup> Edn): 272 | DM Vasudevan (9<sup>th</sup> Edn): 272-275

6. Mechanisms of hormone action.

Satyanarayana (5<sup>th</sup> Edn): 430 | Harper (31<sup>st</sup> Edn): 484 | DM Vasudevan (9<sup>th</sup> Edn): 667, 668

7. Renal Function Tests.

Satyanarayana (5<sup>th</sup> Edn): 460 | Harper (31<sup>st</sup> Edn): 566 | DM Vasudevan (9<sup>th</sup> Edn): 410

### WRITE BRIEFLY ON

[5 × 2 = 10]

8. Define denaturation of proteins. Give examples of denaturing agents.

Satyanarayana (5<sup>th</sup> Edn): 65, 80, 92 | Harper (31<sup>st</sup> Edn): 41 | DM Vasudevan (9<sup>th</sup> Edn): 43

9. What are plasma proteins and write their normal ranges?

Satyanarayana (5<sup>th</sup> Edn): 182 | Harper (31<sup>st</sup> Edn): 113 | DM Vasudevan (9<sup>th</sup> Edn): 425

10. What are oncogenes and give examples?

Satyanarayana (5<sup>th</sup> Edn): 659 | Harper (31<sup>st</sup> Edn): 685 | DM Vasudevan (9<sup>th</sup> Edn): 711

11. Orotic acidurias.

Satyanarayana (5<sup>th</sup> Edn): 402 | Harper (31<sup>st</sup> Edn): 336 | DM Vasudevan (9<sup>th</sup> Edn): 587

12. Mention of biochemical defects in:

(a) Alkaptonuria

(b) Maple syrup urino disease.

Satyanarayana (5<sup>th</sup> Edn): 333, 363 | Harper (31<sup>st</sup> Edn): 286, 290, 291 | DM Vasudevan (9<sup>th</sup> Edn): 311, 400

**M.B.B.S DEGREE EXAMINATION**

July - 2010

**First M.B.B.S Examination****BIOCHEMISTRY****PAPER - I****M.B.B.S DEGREE EXAMINATION**

July - 2010

**First M.B.B.S Examination****BIOCHEMISTRY****PAPER - II****Time: 2½ Hours****Note : Answer all questions : Illustrate your answer with suitable diagrams.****Max. Marks: 50**

1. Explain the formation, utilization and excretion of ketone bodies. What is ketoacidosis? [4 + 2 + 2 + 2 = 10]

Sayanarayana (5<sup>th</sup> Edn): 207, 481 | Harper (31<sup>st</sup> Edn): 207, 210 | DM Vasudevan (9<sup>th</sup> Edn): 213, 215

2. How is acetyl CoA oxidized in citric acid cycle? [3 + 7 = 10]

Sayanarayana (5<sup>th</sup> Edn): 233 | Harper (31<sup>st</sup> Edn): 161 | DM Vasudevan (9<sup>th</sup> Edn): 334**WRITE SHORT NOTE ON** [5 × 4 = 20]

3. Isoenzymes.

Sayanarayana (5<sup>th</sup> Edn): 112, 114 | Harper (31<sup>st</sup> Edn): 61-62 | DM Vasudevan (9<sup>th</sup> Edn): 72

4. Fatty liver.

Sayanarayana (5<sup>th</sup> Edn): 324 | Harper (31<sup>st</sup> Edn): 243, 244 | DM Vasudevan (9<sup>th</sup> Edn): 211, 212

5. Inhibitors of respiratory chain.

Sayanarayana (5<sup>th</sup> Edn): 232 | Harper (31<sup>st</sup> Edn): 118 | DM Vasudevan (9<sup>th</sup> Edn): 352-354

6. Diagnostic importance of enzymes.

Sayanarayana (5<sup>th</sup> Edn): 106 | Harper (31<sup>st</sup> Edn): 63-65 | DM Vasudevan (9<sup>th</sup> Edn): 76, 77

7. Biochemical functions of vitamin D.

Sayanarayana (5<sup>th</sup> Edn): 127 | Harper (31<sup>st</sup> Edn): 528 | DM Vasudevan (9<sup>th</sup> Edn): 492-496**WRITE BRIEFLY ON** [5 × 2 = 10]

8. Cori cycle.

Sayanarayana (5<sup>th</sup> Edn): 261 | Harper (31<sup>st</sup> Edn): 177 | DM Vasudevan (9<sup>th</sup> Edn): 137

9. Hyaluronic acid and its functions.

Sayanarayana (5<sup>th</sup> Edn): 24 | Harper (31<sup>st</sup> Edn): 147, 599, 601 | DM Vasudevan (9<sup>th</sup> Edn): 95, 96

10. What is substrate level phosphorylation? Give two examples.

Sayanarayana (5<sup>th</sup> Edn): 222 | Harper (31<sup>st</sup> Edn): 122 | DM Vasudevan (9<sup>th</sup> Edn): 342

11. Functions of cholesterol.

Sayanarayana (5<sup>th</sup> Edn): 310 | Harper (31<sup>st</sup> Edn): 202 | DM Vasudevan (9<sup>th</sup> Edn): 221

12. What are lyases? Give two examples.

Sayanarayana (5<sup>th</sup> Edn): 89 | Harper (31<sup>st</sup> Edn): 57 | DM Vasudevan (9<sup>th</sup> Edn): 55

1. How is glycine degraded in the body? Outline the synthesis of creatine from glycine. Enumerate the metabolic diseases of glycine. [4 + 3 + 3 = 10]

Sayanarayana (5<sup>th</sup> Edn): 208 | Harper (31<sup>st</sup> Edn): 265, 298, 282-283 | DM Vasudevan (9<sup>th</sup> Edn): 281, 284, 285

2. Explain transcription. Name the different types of RNA and indicate their functions. [6 + 2 + 2 = 10]

Sayanarayana (5<sup>th</sup> Edn): 539 | Harper (31<sup>st</sup> Edn): 341 | DM Vasudevan (9<sup>th</sup> Edn): 608-613**WRITE SHORT NOTE ON** [5 × 4 = 20]

3. Biochemical functions of zinc and iodine.

Sayanarayana (5<sup>th</sup> Edn): 421, 420 | Harper (31<sup>st</sup> Edn): 98, 99 | DM Vasudevan (9<sup>th</sup> Edn): 542

4. Nitrogen balance.

Sayanarayana (5<sup>th</sup> Edn): 509 | Harper (31<sup>st</sup> Edn): 525-526

5. Blood buffers and regulation of pH.

Sayanarayana (5<sup>th</sup> Edn): 475 | Harper (31<sup>st</sup> Edn): 12-13 | DM Vasudevan (9<sup>th</sup> Edn): 441, 442

6. Biologically active peptides.

Sayanarayana (5<sup>th</sup> Edn): 56, 68 | Harper (31<sup>st</sup> Edn): 21, 22

7. Recombinant DNA and its applications.

Sayanarayana (5<sup>th</sup> Edn): 571 | Harper (31<sup>st</sup> Edn): 439 | DM Vasudevan (9<sup>th</sup> Edn): 643, 647**WRITE BRIEFLY ON** [5 × 2 = 10]

8. What is isoelectric pH of a protein? Mention two properties of a protein at this pH.

Sayanarayana (5<sup>th</sup> Edn): 64, 691 | Harper (31<sup>st</sup> Edn): 19 | DM Vasudevan (9<sup>th</sup> Edn): 42

9. What is creatine clearance? What is its diagnostic importance?

Sayanarayana (5<sup>th</sup> Edn): 462 | Harper (31<sup>st</sup> Edn): 566 | DM Vasudevan (9<sup>th</sup> Edn): 415

10. Name any two group I hormones. What is their mechanism of action?

Sayanarayana (5<sup>th</sup> Edn): 429 | DM Vasudevan (9<sup>th</sup> Edn): 667

11. Write the functions of immunoglobulin G.

Sayanarayana (5<sup>th</sup> Edn): 187, 575 | Harper (31<sup>st</sup> Edn): 627 | DM Vasudevan (9<sup>th</sup> Edn): 694

12. What are tumor markers? Give two examples.

Sayanarayana (5<sup>th</sup> Edn): 662 | Harper (31<sup>st</sup> Edn): 700-701 | DM Vasudevan (9<sup>th</sup> Edn): 715

**M.B.B.S DEGREE EXAMINATION**

January - 2010

First M.B.B.S Examination

**BIOCHEMISTRY****PAPER - I****N  
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S****M.B.B.S DEGREE EXAMINATION**

January - 2010

First M.B.B.S Examination

**BIOCHEMISTRY****PAPER - II****Max. Marks: 50**

Note : Answer all questions : Illustrate your answer with suitable diagrams.

**Time: 2½ Hours**

1. Outline the reactions of hexose monophosphate shunt pathway. In which tissues this pathway is operative? What is the significance of this pathway? **[6 + 2 + 2 = 10]**  
*Satyanarayana (5<sup>th</sup> Edn): 346 | Harper (31<sup>st</sup> Edn): 130, 131 | DM Vasudevan (9<sup>th</sup> Edn): 147*

2. How is NADH oxidized in the respiratory chain? Indicate the sites of oxidative phosphorylation. What is chemiosmotic theory? **[4 + 3 + 3 = 10]**  
*Satyanarayana (5<sup>th</sup> Edn): 226, 228, 229 | Harper (31<sup>st</sup> Edn): 118-121 | DM Vasudevan (9<sup>th</sup> Edn): 347, 348*

**WRITE SHORT NOTE ON**

3. Functions and deficiency of Niacin. **[5 × 4 = 20]**  
*Satyanarayana (5<sup>th</sup> Edn): 142 | Harper (31<sup>st</sup> Edn): 534-535 | DM Vasudevan (9<sup>th</sup> Edn): 505, 507*

4. Pyruvate dehydrogenase complex.  
*Satyanarayana (5<sup>th</sup> Edn): 462 | Harper (31<sup>st</sup> Edn): 161, 162 | DM Vasudevan (9<sup>th</sup> Edn): 138*

5. Phospholipids and their functions.  
*Satyanarayana (5<sup>th</sup> Edn): 347 | Harper (31<sup>st</sup> Edn): 196 | DM Vasudevan (9<sup>th</sup> Edn): 104, 105*

6. Glycogen storage diseases.  
*Satyanarayana (5<sup>th</sup> Edn): 269 | Harper (31<sup>st</sup> Edn): 164, 166 | DM Vasudevan (9<sup>th</sup> Edn): 147*

7. Detoxification by conjugation.  
*Satyanarayana (5<sup>th</sup> Edn): 614 | Harper (31<sup>st</sup> Edn): 558 | DM Vasudevan (9<sup>th</sup> Edn): 480*

**WRITE BRIEFLY ON**

8. Functions of high density lipoprotein. **[5 × 2 = 10]**  
*Satyanarayana (5<sup>th</sup> Edn): 317 | Harper (31<sup>st</sup> Edn): 258-259 | DM Vasudevan (9<sup>th</sup> Edn): 230*

9. Biochemically important compounds derived from cholesterol.  
*Satyanarayana (5<sup>th</sup> Edn): 315 | Harper (31<sup>st</sup> Edn): 502 | DM Vasudevan (9<sup>th</sup> Edn): 224*

10. Proenzymes and their importance.  
*Satyanarayana (5<sup>th</sup> Edn): 102 | Harper (31<sup>st</sup> Edn): 87-88*

11. What are ligases? Give two examples.  
*Satyanarayana (5<sup>th</sup> Edn): 87 | Harper (31<sup>st</sup> Edn): 57 | DM Vasudevan (9<sup>th</sup> Edn): 55*

12. Functions of ascorbic acid.  
*Satyanarayana (5<sup>th</sup> Edn): 132 | Harper (31<sup>st</sup> Edn): 528, 539 | DM Vasudevan (9<sup>th</sup> Edn): 519, 520*

1. How is phenylalanine converted to tyrosine in the body? What is phenylketonuria? Outline the formation of thyroid hormones and catecholamines from tyrosine. **[2 + 2 + 3 + 3 = 10]**  
*Satyanarayana (5<sup>th</sup> Edn): 269, 383 | Harper (31<sup>st</sup> Edn): 286 | DM Vasudevan (9<sup>th</sup> Edn): 306-310, 311*

2. Give an account of Watson-Crick model of DNA. List the differences between DNA and RNA. How is DNA replicated? **[3 + 3 + 2 + 2 = 10]**  
*Satyanarayana (5<sup>th</sup> Edn): 254, 291 | Harper (31<sup>st</sup> Edn): 339, 363 | DM Vasudevan (9<sup>th</sup> Edn): 593, 594, 607, 596-600*

**WRITE SHORT NOTE ON**

3. Biochemical functions of copper. **[5 × 4 = 20]**  
*Satyanarayana (5<sup>th</sup> Edn): 142, 355 | Harper (31<sup>st</sup> Edn): 98-100 | DM Vasudevan (9<sup>th</sup> Edn): 541*

4. Clearance Tests.  
*Satyanarayana (5<sup>th</sup> Edn): 138, 460 | Harper (31<sup>st</sup> Edn): 455 | DM Vasudevan (9<sup>th</sup> Edn): 414*

5. Characteristics of genetic code.  
*Satyanarayana (5<sup>th</sup> Edn): 551 | Harper (31<sup>st</sup> Edn): 338 | DM Vasudevan (9<sup>th</sup> Edn): 615*

6. Paper Electrophoresis.  
*Satyanarayana (5<sup>th</sup> Edn): 725 | Harper (31<sup>st</sup> Edn): 628*

7. Metabolic acidosis.  
*Satyanarayana (5<sup>th</sup> Edn): 481 | Harper (31<sup>st</sup> Edn): 275 | DM Vasudevan (9<sup>th</sup> Edn): 446*

**WRITE BRIEFLY ON**

8. Definition and forces responsible for tertiary structure of a protein. **[5 × 2 = 10]**  
*Satyanarayana (5<sup>th</sup> Edn): 58 | Harper (31<sup>st</sup> Edn): 34-41 | DM Vasudevan (9<sup>th</sup> Edn): 37*

9. Name four important substances derived from glycine.  
*Satyanarayana (5<sup>th</sup> Edn): 342 | Harper (31<sup>st</sup> Edn): 298, 300-301 | DM Vasudevan (9<sup>th</sup> Edn): 283*

10. Name any two gastrointestinal hormones. What is their mechanism of action?  
*Satyanarayana (5<sup>th</sup> Edn): 449 | Harper (31<sup>st</sup> Edn): 483, 484 | DM Vasudevan (9<sup>th</sup> Edn): 684, 685*

11. Functions of plasma albumin  
*Satyanarayana (5<sup>th</sup> Edn): 183 | Harper (31<sup>st</sup> Edn): 623, 624 | DM Vasudevan (9<sup>th</sup> Edn): 427*

12. What do you mean by post translational modifications?  
*Satyanarayana (5<sup>th</sup> Edn): 561 | DM Vasudevan (9<sup>th</sup> Edn): 611*

Note : -Answer all questions : Illustrate your answer with suitable diagrams.

Max. Marks: 50

Time: 2½ Hours

1. Discuss the metabolism of ketone bodies and add a note on ketosis. Write briefly on the test done for the detection of ketone bodies in urine. [7 + 3 = 10]  
Sanyalrayana (5<sup>th</sup> Edn): 295, 386 | Harper (31<sup>st</sup> Edn): 207, 210 | DM Vasudevan (9<sup>th</sup> Edn): 213, 215

2. Write the steps of HMP SHUNT pathway. Write the significance of this pathway. [7 + 3 = 10]  
Sanyalrayana (5<sup>th</sup> Edn): 269, 383 | Harper (31<sup>st</sup> Edn): 182, 191 | DM Vasudevan (9<sup>th</sup> Edn): 147

#### WRITE SHORT NOTE ON [5 × 4 = 20]

3. Specificity of enzymes.

Sanyalrayana (5<sup>th</sup> Edn): 98 | Harper (31<sup>st</sup> Edn): 58 | DM Vasudevan (9<sup>th</sup> Edn): 70

4. High energy compounds with example.

Sanyalrayana (5<sup>th</sup> Edn): 220 | DM Vasudevan (9<sup>th</sup> Edn): 345

5. What are porphyrias? Write briefly on acute intermittent porphyria.

Sanyalrayana (5<sup>th</sup> Edn): 210 | Harper (31<sup>st</sup> Edn): 305, 310 | DM Vasudevan (9<sup>th</sup> Edn): 363

6. Functions and deficiency manifestations of vitamin A.

Sanyalrayana (5<sup>th</sup> Edn): 122 | Harper (31<sup>st</sup> Edn): 528, 530 | DM Vasudevan (9<sup>th</sup> Edn): 489, 491

7. Fatty acid synthase complex.

Sanyalrayana (5<sup>th</sup> Edn): 299 | Harper (31<sup>st</sup> Edn): 224, 225 | DM Vasudevan (9<sup>th</sup> Edn): 206

#### WRITE BRIEFLY ON [5 × 2 = 10]

8. Name the enzyme deficit in the following conditions

(a) Galactosemia

(b) Niemann Pick's Disease.

Sanyalrayana (5<sup>th</sup> Edn): 278, 306 | Harper (31<sup>st</sup> Edn): 196, 257 | DM Vasudevan (9<sup>th</sup> Edn): 188, 259, 260

9. Specific Dynamic Action.

Sanyalrayana (5<sup>th</sup> Edn): 503 | Harper (31<sup>st</sup> Edn): 156 | DM Vasudevan (9<sup>th</sup> Edn): 551

10. Anti Vitamins.

Sanyalrayana (5<sup>th</sup> Edn): 95, 162 | DM Vasudevan (9<sup>th</sup> Edn): 569

11. Structure of sucrose.

Sanyalrayana (5<sup>th</sup> Edn): 19 | Harper (31<sup>st</sup> Edn): 145, 146 | DM Vasudevan (9<sup>th</sup> Edn): 92

12. Mechanisms of Detoxification.

Sanyalrayana (5<sup>th</sup> Edn): 344, 455 | Harper (31<sup>st</sup> Edn): 115-116, 556-558 | DM Vasudevan (9<sup>th</sup> Edn): 479

1. How is ammonia detoxified in the body? Give an account of the urea cycle. Add a note on the enzyme defects in the cycle. [2 + 6 + 2 = 10]  
Sanyalrayana (5<sup>th</sup> Edn): 336 | Harper (31<sup>st</sup> Edn): 131-133, 276-278 | DM Vasudevan (9<sup>th</sup> Edn): 272-275

2. Explain the steps involved in the synthesis of recombinant DNA. Mention the clinical application of this technique. [5 + 5 = 10]  
Sanyalrayana (5<sup>th</sup> Edn): 532 | Harper (31<sup>st</sup> Edn): 361-369 | DM Vasudevan (9<sup>th</sup> Edn): 643-647

#### WRITE SHORT NOTE ON [5 × 4 = 20]

3. Metabolic acidosis.

Sanyalrayana (5<sup>th</sup> Edn): 433 | Harper (31<sup>st</sup> Edn): 275 | DM Vasudevan (9<sup>th</sup> Edn): 446

4. Functions and deficiency manifestations of calcium.

Sanyalrayana (5<sup>th</sup> Edn): 263, 128 | Harper (31<sup>st</sup> Edn): 523 | DM Vasudevan (9<sup>th</sup> Edn): 533, 539

5. Primary structure of proteins.

Sanyalrayana (5<sup>th</sup> Edn): 55, 61 | Harper (31<sup>st</sup> Edn): 23-31 | DM Vasudevan (9<sup>th</sup> Edn): 55

6. Mechanism of action of steroid hormones.

Sanyalrayana (5<sup>th</sup> Edn): 39, 313 | Harper (31<sup>st</sup> Edn): 202 | DM Vasudevan (9<sup>th</sup> Edn): 676, 677

7. Hepatic functional tests.

Sanyalrayana (5<sup>th</sup> Edn): 454 | Harper (31<sup>st</sup> Edn): 565 | DM Vasudevan (9<sup>th</sup> Edn): 587

#### WRITE BRIEFLY ON [5 × 2 = 10]

8. Wilson's Disease.

Sanyalrayana (5<sup>th</sup> Edn): 420 | Harper (31<sup>st</sup> Edn): 288, 475 | DM Vasudevan (9<sup>th</sup> Edn): 458, 541

9. Write the normal serum levels of

(a) Sodium

(b) Potassium

(c) Creatinine

(d) Urea.

Sanyalrayana (5<sup>th</sup> Edn): 411, 412, 344, 337 | DM Vasudevan (9<sup>th</sup> Edn): 461, 462, 416, 417

10. Hartnup's disease.

Sanyalrayana (5<sup>th</sup> Edn): 581, 623 | Harper (31<sup>st</sup> Edn): 288, 335 | DM Vasudevan (9<sup>th</sup> Edn): 515

11. Anticancer agents.

Sanyalrayana (5<sup>th</sup> Edn): 664 | Harper (31<sup>st</sup> Edn): 701-7023 | DM Vasudevan (9<sup>th</sup> Edn): 716

12. Write the general structure of immunoglobulins.

Sanyalrayana (5<sup>th</sup> Edn): 187, 375 | Harper (31<sup>st</sup> Edn): 663-664 | DM Vasudevan (9<sup>th</sup> Edn): 683